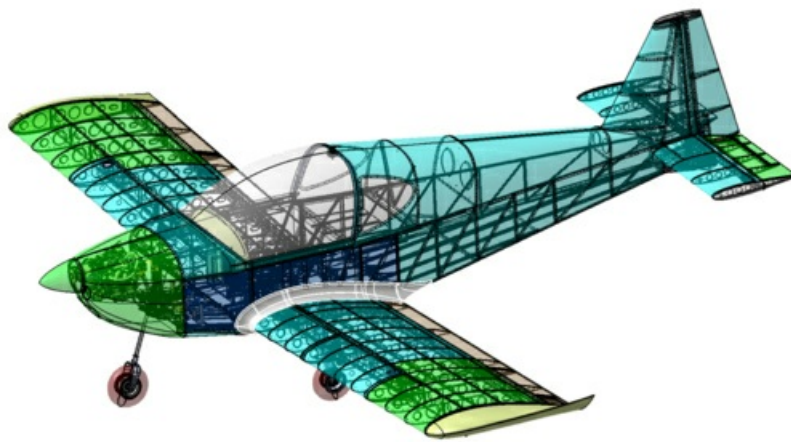


DIRECT FLY ALTO 912 TG AIRCRAFT MAINTENANCE MANUAL



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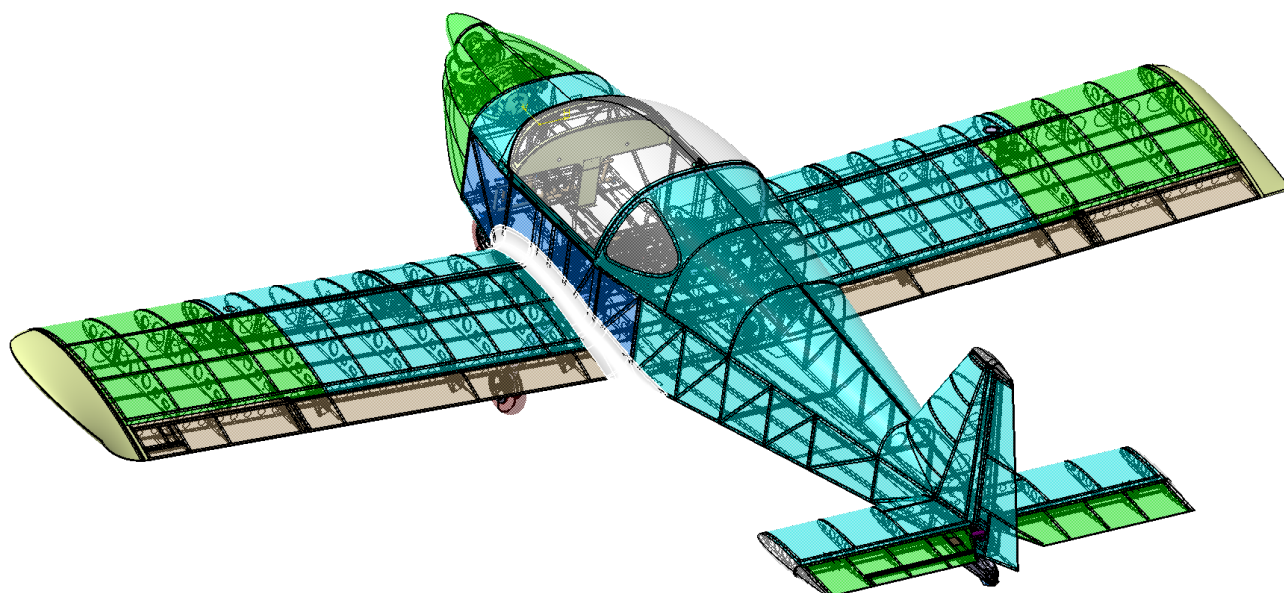


Direct Fly / ERRES

ALTO 912 TG

AIRCRAFT MAINTENANCE MANUAL

DFE-AMM-AR1



Aircraft model / modification: **ALTO****912 TG**

Manufacturer :

Direct Fly s.r.o.

Airplane Serial Number:

DF051

Airplane Registration Number:

PU-RJR

Issued on :

July 24, 2014

Revision :

1

Owner:

Section 0 - INTRODUCTION

0.1. CONTACT INFORMATION

Company headquarters:

Direct Fly s.r. o.
Špitálka 8
602 00 Brno,
Czech Republic
www.directfly.cz
info@directfly.cz

The Manufacturing Facility:

Direct Fly s.r.o.
Cihelní 573
687 25 Hluk
Czech Republic

Brazilian Dealer:

Erres Industria Aeronautica Ltda.
Rua Baden Powel, 43 - Bairro São José
Santa Maria – RS – 79110-120
www.erresaeronautica.com.br
contato@erresaeronautica.com.br

0.2. RECORD OF MANUAL REVISIONS

Refer to Section 12 of this Aircraft Maintenance Manual.

0.3. LIST OF EFFECTIVE PAGES

Refer to Section 12 of this Aircraft Maintenance Manual.

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0.5. SCOPE

This manual has been written by Direct Fly S.R.O. in accordance with the standards of ASTM F2483-12.

The manual provides the practices for the servicing and the maintenance of ALTO 912 TG (LSA) and guidance for the qualifications necessary to perform the various levels of maintenance. It provides practices and guidance for servicing and maintenance that can be performed by a qualified pilot and owner, the preventive maintenance requirements of a 100-hour and an annual condition inspection, and the corrective line maintenance actions for the repairs, alterations, and the removal and re-installation of components.

The owner is reminded that it is her/his responsibility to ensure that Direct Fly S.R.O. has the appropriate contact information so that flight safety and other important information can be communicated in a timely manner. Please use the form on Paragraph 0.10 and email or fax to the addresses listed on Paragraph 0.1 on this Section.

NOTE

Direct Fly S.R.O. cannot through this manual address all of the safety concerns associated with the use of this document. It is the responsibility of the user of this document to establish appropriate safety and health practices and to determine the applicability of any regulatory limitations prior to use.

0.6. PILOT / OWNER SERVICING AND MAINTENANCE

This manual identifies servicing and maintenance actions, which can be performed by the owner/pilot, as, identified in ASTM F2483, and who holds a light sport aircraft pilot's certificate (or greater) and who may not have received any specific authorized Direct Fly S.R.O. training. The maintenance actions authorized in this manual comply with the intent of FAA 14 CFR Part 43.3 with regard to preventive maintenance.

0.7. PREVENTIVE MAINTENANCE

This manual identifies the Direct Fly ALTO preventive maintenance tasks, which are to be accomplished by qualified personnel, and include the actions for the 100-hour inspection and the annual inspection.

0.8. CORRECTIVE MAINTENANCE & ALTERATIONS

This manual identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, or in this maintenance manual, are authorized at this time, for this ALTO aircraft.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

0.9. CONTINUED OPERATIONAL SAFETY REPORTING

The owner/operator of a LSA is responsible for notifying the manufacturer of any safety of flight issue or significant service difficulty upon discovery. The form in Section 13 of this Manual may be used & sent to Direct Fly S.R.O by email or fax to the addresses listed on Paragraph 0.1 on this Section

0.10. OWNER CHANGE OF ADDRESS NOTICE

Each owner/operator of a LSA is responsible for providing the manufacturer with current contact information where the manufacturer may send the owner/operator supplemental notification bulletins. The following form may be used & sent to Direct Fly S.R.O by email or fax to the addresses listed on Paragraph 0.1 on this Section

CHANGE OF OWNERSHIP OR ADDRESS			
Aircraft Model			
Aircraft Registration			
Aircraft Serial Number			
Owner			
Address			
e-mail		Phone	
Signature			
NEW CONTACT INFORMATION			
Owner			
Address			
e-mail		Phone	
Signature			

0.11. TERMILONOLOGY

A&P: A US FAA certificated Mechanic as defined by 14 CFR Part 65.

LSA Repairman Inspection: A US FAA certificated repairman (light-sport aircraft) with and inspection rating as defined by 14 CFR part 65, authorized to perform the annual condition inspection on light-sport aircraft, which the repairman owns.

LSA Repairman Maintenance: A US FAA certificated repairman (light-sport aircraft) with a maintenance rating as defined by 14 CFR part 65, authorized to perform line maintenance on aircraft certificated as special LSA aircraft.

LSA Aircraft (Light Sport Aircraft) - An aircraft designed per ASTM F2245 (airplanes) and complies with the U.S. Code of Federal Regulations Chapter 14 Part 1 definition for a light sport aircraft and will be certificated in the US as SLSA or ELSA.

FAA: United States Federal Aviation Administration

14 CFR: Code of federal regulations chapter 14 Aeronautics and Space also know as the "FARs" or federal aviation regulations.

Manufacturer – Any entity engaged in the production of a SLSA or ELSA or component used on an LSA.

Maintenance Manual(s) – A manual provided by a SLSA or ELSA manufacturer (this manual) or supplier that specifies all maintenance, repairs and alterations authorized by the manufacturer.

Line Maintenance – Any unscheduled maintenance resulting from unforeseen events, or scheduled checks that contain servicing and/or inspections not considered heavy maintenance that is approved by the manufacturer and is specified in the manufacturer's maintenance manual.

Typical tasks considered as line maintenance for LSA 's include:

1. 100 hour inspection
2. Annual Condition Inspection
3. Servicing of fluids
4. Removal and replacement of components for which instructions are provided in the maintenance manual
5. Repair of components and structure for which instructions are provided in the maintenance manual and which do not require additional specialized training such as:
6. Compliance with a manufacturer service directive when the repairman is listed as an authorized person to accomplish the work described.

Heavy Maintenance – Any maintenance, inspection, repair or alteration that requires specialized training, equipment, or facilities.

Authorization to perform: The holder of an FAA Mechanic certificate with Airframe and or Powerplant rating(s); or a LSA Repairman maintenance that has received additional task specific training for the function to be performed is generally considered the minimum level of certification to perform heavy maintenance of SLSA in the U.S..

Typical tasks considered as heavy maintenance for SLSA 's include:

1. Removal and replacement of components for which instructions are provided in the maintenance manual or service directive instructions such as:
 - Complete engine removal and reinstallation in support of an engine overhaul or to install a new engine.
 - Primary flight control cables/components
 - Landing gear assemblies
2. Repair of components and or aircraft structure for which instructions are provided in the maintenance manual or service directive instructions such as:

Repainting of control surfaces
Structural repairs

3. Alterations of components and or aircraft structure which instructions are provided in the maintenance manual or service directive instruction such as:
Installation of new equipment authorized by the aircraft manufacturer

Overhaul – Maintenance, inspection, repair or alterations that are only to be accomplished by the original manufacturer or a facility approved by the original manufacture of the product.

Authorization to perform: Only the manufacturer of a SLSA or the component to be overhauled on a SLSA may perform or authorize to be performed the overhaul of a SLSA or SLSA component. No FAA certification is given to be a SLSA approved overhaul facility.

Overhaul manual: A separate overhaul manual, not the manufacturer's maintenance manual, is required to perform the overhaul of a SLSA or SLSA component. The form and content of such a manual is not governed by this standard or by any FAA regulation. Note: Specific form and content guidelines have not been promulgated here as type specific training and authorization is required from the manufacturer in order to overhaul a SLSA or component.

Typical components that are overhauled include:

Engines
Carburetors/fuel injection systems
Starters/alternators/generators
Instruments

Overhaul Facility – A facility specifically authorized by the aircraft or component manufacturer to overhaul the product originally produced by that manufacturer.

Annual Condition Inspection – A detailed inspection accomplished once a year on a SLSA or ELSA in accordance with instructions provided in the maintenance manual (refer to 2.4.3. Periodical inspection intervals) supplied with the aircraft. The purpose of the inspection is to look for any wear, corrosion, or damage that would cause an aircraft to not be in a condition for safe operation.

100 Hour Inspection – Same as an annual condition inspection except the interval of inspection is 100 hours of operation instead of 12 calendar months. This inspection is utilized when the SLSA or ELSA is being used for commercial operations such as flight instruction and or rental.

0.12. QUALIFICATION REQUIREMENTS

Maintenance and Repairs of LSA

The factory-built Alto belongs to Special-Light Sport Aircraft category, while the Alto built from a kit to Experimental-Light Sport Aircraft category.

This fact determines the conditions for aircraft maintenance and repairs, as explained hereafter:

REVIEW OF THE REQUIRED QUALIFICATION FOR MAINTENANCE AND INSPECTIONS:

	Alto built from a kit (ELSA under §21.191 (i) 1,320 pounds MTOW)	Factory-built Alto (SLSA under §21.190 1,320 pounds MTOW)
Preventive maintenance	Sport pilot or higher	Sport pilot or higher
Maintenance	Owner-maintained	- Repairman (LS-M) - Mechanic (A&P) - Part 145 Repair station
Inspections	Annual condition - Repairman LS –I and LS-M - Mechanic (A & P) - Part 145 Repair station 100-hour condition - Repairman LS-M - Mechanic (A & P) - Part 145 Repair station	Annual condition: - Repairman LS –I and LS-M - Mechanic (A & P) - Part 145 Repair station 100-hour condition - Repairman LS-M - Mechanic (A & P) - Part 145 Repair station

REVIEW MAINTENANCE TYPE VERSUS REQUIRED QUALIFICATION

LINE MAINTENANCE	HEAVY MAINTENANCE	OVERHAUL
Authorization to perform: - Aircraft owner (only for ELSA) - Repairman (LS-M) - Mechanic (A&P) - Part 145 Repair station * Repairman LS –I and LS-M, Mechanic (A&P) Part 145 Repair station (For Annual Condition Inspection)	Authorization to perform: - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.. - FAA approved Part 145 Repair station	Authorization to perform: - manufacturer of a SLSA / ELSA or the component - facility specifically authorized by the aircraft or component manufacturer
Typical tasks 100 hour inspection *. Annual Condition Inspection - Servicing of fluids - Removal and replacement of components for which instructions are provided in the maintenance manual - Repair of components and structure for which instructions are provided in the maintenance manual and which do not require additional specialized training. - Alterations for which specific instruction are provided in the maintenance manual. - Compliance with a manufacturer service directive.	Typical tasks: - Removal and replacement of components for which instructions are provided in the maintenance manual or service directive instructions such as: - Complete engine removal and reinstallation in support of an engine overhaul or to install a new engine. - Primary flight control cables/components - Landing gear assemblies - Repair of components and or aircraft structure for which instructions are provided in the maintenance manual or service directive instructions such as: - Repainting of control surfaces - Structural repairs - Alterations of components and or aircraft structure which instruction are provided in the maintenance manual or service directive instruction such as: - Installation of new additional equipment listed as approved equipment by the aircraft manufacture	Typical tasks: Overhaul of the: - Engines - Carburetors/fuel injection systems - Starters/alternators/generators - Instruments

0.12. SIGNIFICANCE AND USE

The purpose of this maintenance manual is to provide guidance to owners, mechanics, airports, regulatory officials, and aircraft and component manufacturers who may accomplish maintenance, repairs, and alterations on the Direct Fly ALTO Aircraft.

Section 1 – GENERAL INFORMATION

1.1. GENERAL

Direct Fly S.R.O. prepared this aircraft maintenance manual. It contains the information that a trained mechanic will need to maintain the aircraft in an airworthy condition.

The Aircraft Maintenance Manual was prepared to meet the ASTM F2483 Standard Specification requirements. The format of the manual is designed to be familiar to most modern aircraft mechanics.

This maintenance manual does not reflect part numbers and cannot be used for ordering replacement parts.

The wiring schematics that have been included in the manual are for general information purposes only. Aircraft will have optional equipment for which there will be specific drawings that will either be included with the aircraft's documentation on delivery.

1.2. LAYOUT OF THE MANUAL

This maintenance manual is divided into Sections. The Section numbers appear at the foot of each page.

1.3. WARNINGS, CAUTION AND NOTES WARNING

WARNING!

AN OPERATING PROCEDURE, PRACTICE, OR A CONDITION, WHICH, IF NOT CORRECTLY FOLLOWED OR REMEDIED, COULD RESULT IN SERIOUS PERSONAL INJURY OR LOSS OF LIFE.

CAUTION

An operating procedure, practice, or a condition, which, if not strictly observed or corrected, could result in destruction of, or damage to equipment.

NOTE

An operating procedure, practice or condition which is important to emphasize.

1.4. SPECIFICATION

Direct Fly ALTO 912 TG is an all-metal low wing aircraft with a riveted aluminum structure. Some non-load bearing parts such as the engine cowling, wing tips, empennage tips, and wheel covers are made of composite materials. The Alto is powered by the ROTAX 912 ULS engine (100 HP) and PROPULS 174/3/R three-blade ground-adjustable propeller. The two seats are arranged in a side-by-side configuration. The tricycle landing gear features a steerable nose wheel and hydraulic brakes on main wheels.

1.4.1. Basic Technical Data

1.4.1.1. Side, Top, and Front Views.

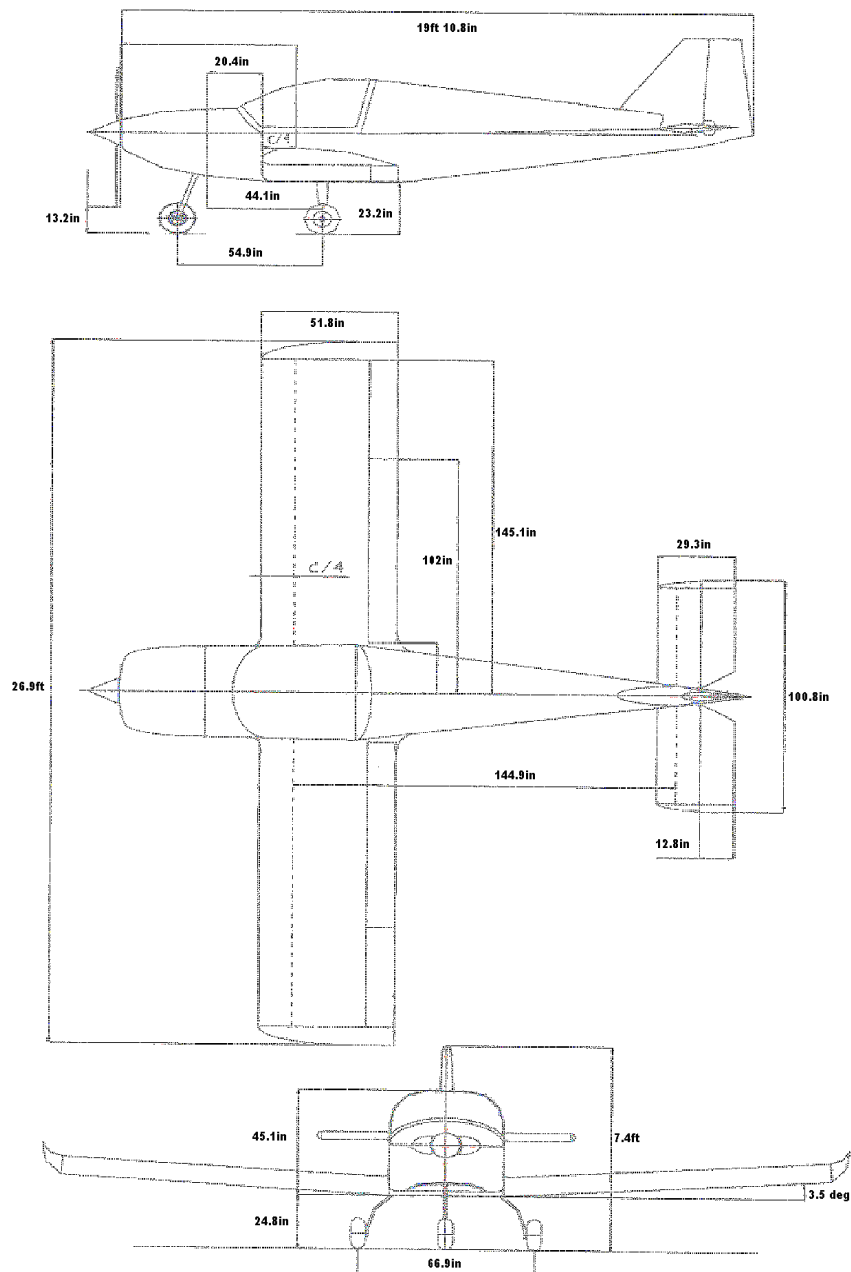


Fig 1-1: Side, Top, and Front View

1.4.1.2. Basic dimensions**Wing**

Span	322.83 in	8.20 m
Area	114 sq.ft	10.52 sq.m
MAC	51.77 in	1.315 m
Loading	11.71 lbs/sq.ft	59.18 kg/sq.m

Aileron

Area	2.62 sq.ft	0.25 sq.m
------------	------------	-----------

Flap

Area	5.60 sq.ft	0.52 sq.m
------------	------------	-----------

Fuselage

Length	20.18 ft	6.15 m
Width	3.61 ft	1.10 m
Height	7.36 ft	2.246 m
Cockpit canopy max. width	3.9 ft	1.180 m

HTU

Span	8.4 ft....	2.56 m
HTU area	20.88 sq.ft	1.768 sq.m
Elevator area	8.4 sq.ft	0.8 sq.m

VTU

Height	4.21 ft..	1.28 m
VTU area	10.99 sq.ft	1.03 sq.m
Rudder area	4.67 sq.ft	0.44 sq.m

Landing gear

Wheel track	5.57 ft..	1.70 m
Wheel base	4.57 ft..	1.395 m
Main wheel diameter	15 in....	380 mm
Nose wheel diameter	15 in	380 mm

1.4.1.3. Weight

Empty weight (This is an average wt. Actual wt. is in POH)...	700 lbs	317 kg
Maximum Take-off weight	1320 lbs	600 kg
Maximum Landing weight	1320 lbs	600 kg
Maximum weight in Baggage Compartment	33 lbs	15 kg

NOTE

An aircraft's actual empty weight will vary based on how it is equipped.
Each aircraft's empty weight is listed in its Pilot Operating Handbook

1.4.1.4. Center of gravity

Empty weight CG (standard equipment).....21-25% MAC
Operating CG 25-35% MAC
(MAC..... Mean Aerodynamic Chord)

1.4.1.5. Operating limitations

Refer to the PILOT'S OPERATING HANDBOOK - POH - Section 2 for more details about the following operating limits:

- Airspeed limits
- Weight limits
- CG Range limits
- Approved maneuvers

NOTE

Additional rules are of a more common character and result from generally valid flight regulations.
It is in every user's interest to be familiar with these regulations, rules and restrictions.

1.5. GROUND HANDLING

All ground handling activities described below are considered as Line Maintenance and may be performed by Sport Pilot or higher (or Aircraft owner for ELSA).

1.5.1. Towing

It is easy to tow the airplane a short distance by holding the propeller blade root, since the empty weight of this airplane is relatively low.

Suitable surfaces to hold the aircraft airframe are the rear part of the fuselage before the fin and the wing roots.

A tow bar may be used to tow the aircraft over long distances. The steerable nose wheel is equipped with stops, to limit its travel.

1.5.2. Parking the airplane

It is advisable to park the aircraft inside a hangar, or eventually inside a other weather proof space (such as a garage) with stable temperature, good ventilation, low humidity and a dust-free environment.

It is necessary to tie-down the aircraft when parking outside.

On occasions when the plane must be tied-down outdoors for extended periods, it is advisable to cover the cockpit canopy, and if possible, the entire aircraft using a suitable cover.

We recommend removing the battery in winter and storing it at indoor temperature.

CAUTION

Avoid parking airplane (especially airplane with dark design) on the direct sunshine during warm climate - then "can effect" or wave can arise on the upper skin of the wing and fuselage. It disappears when airplane is parked in the hangar, in the shadow or during taxiing or in flight when airplane surface is cooled with air flow. This "can effect" or wave arise during parking has no effect onto functionality and it cannot be subject of claim.

1.5.3. Tie-Down

The airplane is usually tied-down when parked outside a hangar. The tie-down is necessary to protect the aircraft against possible damage caused by wind gusts.

For this reason the aircraft is equipped with tie-down eyes, located on the wing lower surface.

Procedure:

Check: Fuel valve off, Master switch and ignition switches off.

Block the control stick up e.g. by means of safety harness or connect the control stick with rudder pedals by means of a suitable rope.

Shut all the ventilation windows

Close and lock cockpit

Tie-down the aircraft to the ground by means of a mooring rope passed through the eyes, located on the wing lower surface. It is also necessary to tie the nose wheel landing gear to the ground.

NOTE

It is advisable to cover cockpit canopy, if possible the whole airplane, by means of a suitable covering material attached to the airframe for long term outside parking

1.5.4. Jacking the airplane

Because the empty weight of this airplane is relatively low it is easy to lift the airplane using 2 people.

First prepare two suitable jacks or supports to support the aircraft. Aircraft fuselage can be supported on the places marked by the black triangles in the figure below.

The aircraft should be lifted by the following parts:

- Press-down on the rear of the fuselage in front of the fin to lift the front and then support under the firewall.
- To lift the wings, push on the wings lower surface at the main spar. Do not lift by the wing tips.

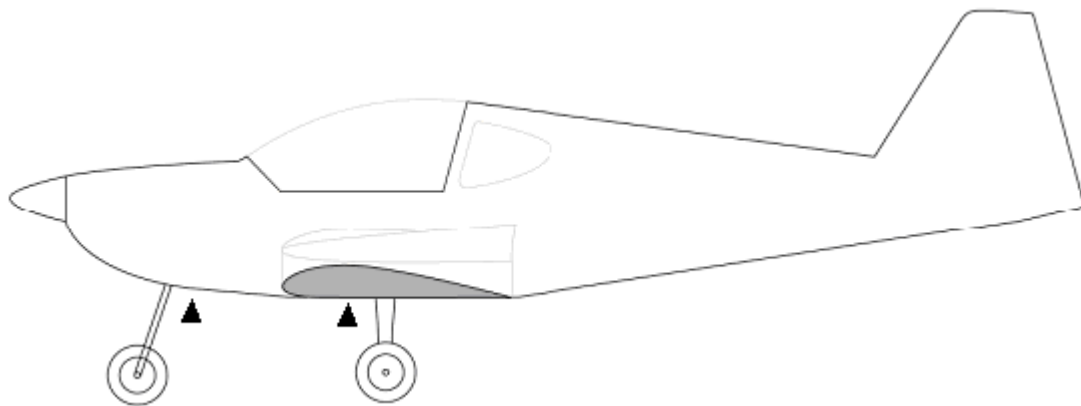


Fig 1-2: Lifting support places

1.6. SERVICING

1.6.1. Fueling

WARNING!

No smoking or open flames during fueling!
Fire extinguisher should be within reach!
Under no circumstances add fuel with the engine running!
Connect the aircraft to ground prior fueling.
No person in the cockpit during fueling!

A fuel tank filler is located on the upper side of each wing (see figure below).



Fig 1-3: Fuel tank filler Cap

- Tools needed: None
- Parts needed: None
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions:
 1. Connect the aircraft to the ground.
 2. Open fuel tank filler
 3. Insert fuel delivery nozzle or a funnel into the filler.
 4. Fill required quantity of fuel.
 5. Remove nozzle.
 6. Close fuel tank filler by its cap.

- Task proper accomplishment check: check the fuel tank cap is home screwed and required amount of fuel is in the tank (according to fuel gauge).

CAUTION

It is highly recommended to pour gasoline through a filter if it was not tested for water content. After fueling, allow 20 min. for water to settle out on the bottom. Drain off some fuel and look for water.

1.6.1.1. Approved Fuel Grades and Specifications

- Min. RON 95, EN 228 Premium, EN 228 Premium plus or AVGAS100LL
- Fuel according to FAA - Standard Spec. for Automotive Spark-Ignition Engine Fuel, ASTM D 4814 or AVGAS 100 LL.
- Fuel according to DOT - CAN/CGSB-3.5 Quality 3 min AKI 91 or AVGAS 100 LL.

Refer to the BRP-ROTAX engine Operator's Manual - Section 2.4

1.6.2. Oil Filling

The oil reservoir is located in the engine compartment and is accessible when engine upper cowling is removed. Oil quantity is measured by wire-gauge under the Oil Reservoir Cap (see photo).



Fig 1-4: Oil Reservoir

- Tools needed: Screwdriver to remove upper cowling
- Parts needed: recommended brand of oil
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions:

1. Remove upper cowling.
2. Unscrew oil reservoir cap.
3. Verify the oil level.

The total oil quantity in the Rotax 912 ULS lubricating system amounts to 0.9 U.S. gall (3.5 liters). Prior to oil check, turn the propeller in the normal direction of rotation by hand (ignition switched off!) several times to pump oil from the engine into the oil tank, or leave the engine idle for 1 minute. The oil level in the oil reservoir should be between the min. and max. marks and should not be below min. mark.

4. If the oil level is below min mark, insert a funnel and fill oil.
 5. Remove funnel, screw oil reservoir cap, check cap proper attachment.
 6. Mount upper cowling
- Task proper accomplishment check: check the fuel tank cap is home screwed and required amount of fuel is in the tank (according to fuel gauge).

1.6.2.1 Approved Oil Grades and Specifications

Motor oil tested and released from BRP-Rotax (for use with unleaded fuel or MOGAS).

Shell Sport Plus 4 API SL SAE 10 W-40 Code 2.

Refer to the BRP-ROTAX engine Operator's Manual - Section 2.5

1.6.3. Coolant Filling

The expansion reservoir located in the engine compartment attached to the firewall is used for filling the coolant. In addition to that, this reservoir absorbs coolant in the case of engine overheating. (see figure below).



Fig 1-5: Coolant expansion reservoir

- Tools needed: Screwdriver to remove upper cowling
- Parts needed: recommended coolant
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions:
 1. Remove upper engine cowling.
 2. Check coolant quantity
The total coolant quantity is about 0.4 U.S. gal (1.5 liters).
 3. Unscrew coolant expansion reservoir cap.
 4. fill some amount if needed.
 5. Screw tank cap back
 6. Mount upper cowling back.
- Task proper accomplishment check: Check proper attachment of coolant tank cap after refilling.

1.6.3.1 Recommended Types of Coolant

The "BASF Glysantin Protect Plus/G48" is recommended by the engine manufacturer. The engine manufacturer also recommends the use of antifreeze concentrate during cold weather operation. Castrol Radicool NF can also be used.

Refer to the BRP-ROTAX engine Operator's Manual - Section 2.3

1.6.4. Brake Fluid Filling

A brake fluid is filled into reservoir located below the central panel between the seats - see "C" on the picture below. For the minimum and maximum brake fluid level – see picture below.

Brake fluid refilling is necessary when a low brake system efficiency occurs due to a fluid leak.

- Tools needed: Screwdriver
- Parts needed: recommended brake fluid
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions: See pictures below.
 1. Disconnect the brake grip A
 2. Remove screws B
 3. Open the central panel C
 4. Check brake fluid quantity and fill some amount if needed.
 5. Brake repeatedly on the brake lever during refilling. Bleed the system after refilling.
 6. Screw tank cap back, install panel C and screws B, connect grip A

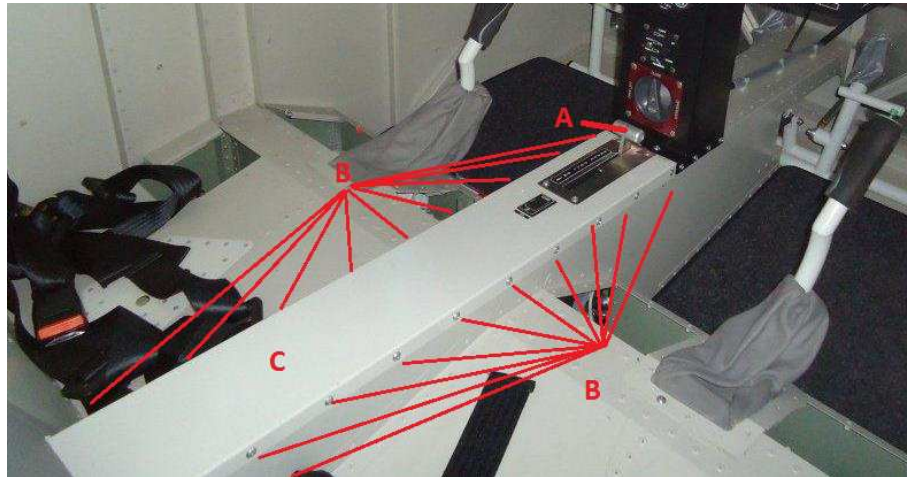


Fig 1-6: Brake Fluid Reservoir Location



Fig 1-7: Brake Fluid Reservoir



Fig 1-8: Brake Fluid Level Marks

- Task proper accomplishment check: Check proper attachment of brake fluid tank cap after refilling.

1.6.4.1 Recommended Type of Brake Fluid

ONLY DOT 4 or DOT 5.1 fluid should be used.

1.6.5. Lubrication

- Tools needed: Screwdriver to remove upper cowling, wheel pants and access windows, grease gun, oiler.
- Parts needed: recommended Grease and Oil
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions:
 1. Remove upper engine cowling, wheel pants and access windows
 2. Lubricate recommended points
 3. Mount upper cowling, wheel pants and access windows back.
- Task proper accomplishment check: Check proper attachment of upper cowling, wheel pants and access windows.

1.6.5.1. Lubrication fundamentals

There are some generally inaccessible joints and control system parts inside the wings and fuselage, which have been cleaned and lubricated during airplane assembly. Lubrication of these will be performed during a periodic inspection.

There are some parts, e.g. landing gear, which are exposed to external conditions and to varying loads. These parts will be inspected during pre-flight and during periodical inspections. These should be lubricated as is necessary, but at least in the intervals specified below.

1.6.5.2. Recommended lubricants

- **Greases**

Greases are mineral oils thickened with calcic, sodium, lithium or any other thickeners of aliphatic acids. Grease may be applied all the year round.

The following greases are recommended:

- waxy, semi-solid or butyraceous consistency and water resistant. They are used at very low temperatures (-22 °F, -30 °C) and at high temperatures (248 °F, 120 °C)

Czech	Foreign
MOGUL MOLYKA G (or equivalent)	AEROSHELL GREASE 22
	AEROSHELL GREASE 11MS
	AEROSHELL GREASE 23C
	SHELL RETINAX HDX2
	SHELL RETINAX EPX2
	(or equivalent)

- **Lubricating Oils**

Use WD-40 or equivalent to lubricate the recommended points.

1.6.5.3. Lubricating Points

Unit	Lubricating point	after the first 25 hrs.	Every 50 hrs.	Every 100 hrs	Lubricant
Engine	• carburetor control cable at inlet into the Bowden (in engine compartment)	x	x		WD40
	• choke control cable at inlet into the termination (in engine compartment)	x	x		WD40
Nose wheel	• landing gear leg in the area of bushing	x	x		MOGUL MOLYKA G, foreign greases
Main landing gear	• pins of brake pads holders		x		MOGUL MOLYKA G, foreign greases
Ailerons	• hinges		x		MOGUL MOLYKA G, foreign greases
	• control hinge pin			x	MOGUL MOLYKA G, foreign greases
	• two-armed aileron control levers inside the wing			x	MOGUL MOLYKA G, foreign greases
	• hinge joint of rods under the wing fillet			x	MOGUL MOLYKA G, foreign greases
Flaps	• hinges	x	x		MOGUL MOLYKA G, foreign greases
	• all movable joints under the quadrant cover between the seats			x	MOGUL MOLYKA G, foreign greases
	• All movable joints under the baggage compartment bottom cover			x	MOGUL MOLYKA G, foreign greases
	• Flaps control pins (at a flap root)		x		MOGUL MOLYKA G, foreign greases
HTU	• elevator hinge		x		MOGUL MOLYKA G, foreign greases
	• swivel bearing in the elevator control rod termination			x	MOGUL MOLYKA G, foreign greases
VTU	• rudder suspensions			x	MOGUL MOLYKA G, foreign greases
	• rudder control cables at attachment to the rudder			x	MOGUL MOLYKA G, foreign greases
Trim tab	• trim tab hinge	x	x		WD40
Stick control	• All movable joints in the cockpit			x	MOGUL MOLYKA G, foreign greases
Rudder control	• All movable joints in the cockpit			x	MOGUL MOLYKA G, foreign greases
	• The passages of rudder control cables			x	MOGUL MOLYKA G, foreign greases
	• Brake master attachment points at rudder peddles			x	MOGUL MOLYKA G, foreign greases

1.6.6. Winter operation

1.6.6.2. Definition

It is considered a winter operation, if the outside temperature falls below 41°F (+5 °C).

1.6.6.2. Preparing the aircraft for winter operation

- Tools needed: Screwdriver to remove upper cowling, wheel pants and access windows, grease gun, oiler.
- Parts needed: recommended Grease, Oil and anti-freezing
- Type of maintenance: Line
- Level of Certification needed: Refer to Section 0 – Paragraph 0.12.
- Instructions:

Aircraft airframe

1. Lubricate the aircraft per Lubricating Chart (100 hr. Inspection) if the last inspection was not within 6 months
2. Check and adjust rudder and elevator control cable “pre-stress”
3. Check cockpit canopy rubber packing – replace if damaged
4. Check fuel tank venting
5. Check attachment of wing, ailerons, flaps and tail units; lubricate per Lubricating Chart

Engine

Refer to the Engine Manual for more details.

The following should be done:

1. Add Anti-freeze to the cooling system as required (usually a 50/50 mix.)
2. Change the oil (see Engine Manual or Service instructions)
3. Check the spark plug gaps

If low cylinder head or oil temperatures occur during operation under low outside temperature, then do the following:

1. Cover a portion of the radiator face using an aluminum sheet, insert it between the radiator and the lower engine cowling.
 2. Cover the oil cooler face (or a part of face) using an aluminum sheet attached with a suitable adhesive tape.
 3. Cover the Reduction gear unit by means of a car engine cover
 4. Cover the oil filter (see instructions for Reduction gear unit)
 5. Cover the oil tank (see instructions for Reduction gear unit)
- Task proper accomplishment check: Check proper attachment of upper cowling, wheel pants and access windows.

1.6.6.3. Additional Preflight inspection

In addition to the Pre-flight inspection described in the POH – Section 4, the following must be done:

- Remove all snow from the airplane surfaces, and remove any icing using hot air
- Check the control surfaces for free movement. Check the slots of the control surfaces and flaps. Remove any snow or ice
- Check the fuel tank vent hose for openness before each flight
- Check fuel system (fuel filter) for debris, if found, empty the fuel tank and refill with fresh fuel
- Check fuel for water prior filling (use of water separator/filter is recommended)
- Drain fuel tank (drain valves are located on the bottom side of the each wing), drain a small quantity of fuel; check for water and other contaminants until fuel is clean and clear.

1.6.6.4. Pre Heating Engine and Oil

It is permissible to start an engine without pre-heating if the outside air temperature is not below +5 °C. Pre-heat the engine and oil if air temperature falls below 41°F (+5 °C). Use suitable air heater or a dryer.

WARNING!

NEVER USE OPEN FIRE TO PRE-HEAT AN ENGINE!

Blow hot air from the front into the hole around the prop (engine covered with fiberglass cowlings).

The temperature of the hot air should not exceed 212°F (100°C) at air heater output. Warm up the oil tank along with the oil in the engine. Pre-heat until cylinder head and oil temperatures exceed 68°F (+20° C).

1.6.6.5. Engine Starting

1. Turn the propeller by hand normal direction of rotation (NEVER turn the engine propeller opposite normal direction of rotation) Refer to Rotax engine information. (ignition switched off!)
2. Set the fuel selector to LEFT position
3. Set throttle lever to idle
4. Open the choke
5. Master switch to "ON"
6. Switch on ignition to "START" after starting to "BOTH"
7. Verify oil pressure and adjust engine RPM after starting
8. Close the choke
9. Warm up the engine

CAUTION

If the cylinder head and oil temperatures fall during parking. Start and warm up engine from time to time between flights. Do not open choke when starting a hot engine.

1.6.6.6. Parking and Taxiing

Check wheel brakes for freezing when parked outside and temperature is below zero. Check wheels free rotation prior to taxiing (Grasp the propeller at the blade shank and pull the airplane). Heat the brakes with hot air (to melt snow or ice). Frozen materials should not be removed by forced towing.

1.6.6.7. After Winter Operation

- Clean the airplane thoroughly (hinges, especially the suspensions)
- Lubricate airplane per Lubricating Chart
- Check and adjust the control system

1.7. EQUIPMENT LIST

Direct Fly S.R.O. retains the approval authority of any items installed in all ALTO aircrafts. Therefore the following master list of equipment must be enforced as the only approved items for installation on the aircraft without further authority. No substitutions are allowed without written approval by the Manufacturer.

In the following table there is a list of components that the owner or operator might change or upgrade (S= Standard, O= Optional).

ITEM	BRAND	MODEL / COMMENT	S/O
Engine and accessories			
Engine	BRP/Rotax	912 ULS	S
Auxiliary Fuel Pump	Pierburg	7,21440,51,0	S
Cabin Heater	Direct Fly		S
Propeller	WoodComp	PROPULS 174/3/R	S
Wheels and Tires			
Front wheel disc	Aerospool	02-A Veral	S
Front wheel tire	Sava or similar	13x5.00-6 B19	S
Main wheel disc	Aerospool	02-A Veral	S
Main wheel tire	Sava or similar	14x4, 4.00-6/6 PR B11	S
Landing Gear			
Main Landing Gear	Comlet	NK03M	S
Nose Landing Gear	Direct Fly	DF	S
Brakes	Aerospool	ASP, MAGURA 012 2020 225.4	S
Wheel Pants	Direct Fly	DF	S
Airframe			
Canopy Key Lock	Direct Fly		S
Canopy	Direct Fly		S
Towing Hook System	Aerotechnik		O
Electric Trim	Ray Allen	T10	S
Strobe Lights	Direct Fly	Kunzelman	S
	SimplePack	Brazilian Version	O
Landing Lights	Direct Fly	LED 9	S
Cowling – Upper Section	Direct Fly		S
Cowling – Lower Section	Direct Fly		S
Avionics / Instruments			
VHF	Icom	IC-A210	S
Transponder	Garmin	GTX-327	S
GPS	Garmin	Aera-500	S
ASI	PRC	BK-15	S
Altimeter	Mikrotechna	LUN 1128	S
Climb	PRC	BC10-1B	S
Compass	PRC	CM-24S	S
Slip Indicator	Winter	WIQM2	S
Tachometer	Road	IF61.2B	S
Hourmeter	Honneywell	85094	S
Oil Pressure	Road	IE61.2B	S
Oil Temperature	Road	IH61.2B	S
CHT	Road	IH71.2B	S
Fuel Level Indicator	VDO	FL314	S
Flap Controller	FlyBox	EFC-P	S
Circuit Breaker	ETA	1110-F212-P1M1- A	S
Master/Start Key	CZ - VDO	VDO	S

Some items in the previous list are described as “Direct Fly”, which does not mean that Direct Fly is the manufacturer.

For their nature, and availability, instruments can be of different brands: the substitution of one instrument for one of another brand but analog technology and specifications, can provide satisfactory results. Even if the change at first sight has no consequence on CG or other aspects, we encourage the owner to take contact with Direct Fly or its dealers before attempting any modification or change in the equipment.

1.8. SOURCES TO PURCHASE PARTS

Spare parts can be ordered from different sources depending on its nature. We list here a simplified list of sources, and we encourage the owner to contact Direct Fly or its Dealers anytime a question arises about the compatibility of a product or consumable.

A separate list for sources for disposable replacement parts can be found in the next paragraph.

Different sources when the source indicated is not Direct Fly, do not necessarily imply incompliance.

Parts from Direct Fly are accessible through the entire dealer network which is kept up to date in the website: <http://www.directfly.cz>

ITEM	SOURCE
Engine and accessories	
Engine Parts	Rotax Dealers / Direct Fly Dealers
Auxiliary Fuel Pump	Direct Fly Dealers or automotive parts dealer
Cabin Heater	Direct Fly Dealers
Propeller	WoodComp Dealers / Direct Fly Dealers
Wheels and Tires	
Front wheel complete	Direct Fly Dealers
Tire + tube for front wheel	Direct Fly Dealers or aircrafts part shop
Main wheels complete	Direct Fly Dealers
Tire + tube for main wheels	Direct Fly Dealers or aircrafts part shop
Landing Gear	
Main Landing Gear	Direct Fly Dealers or Comlet
Nose Landing Gear	Direct Fly Dealers
Brakes	Direct Fly Dealers
Wheel Pants	Direct Fly Dealers
Airframe	
Canopy Key Lock	Direct Fly Dealers
Canopy parts	Direct Fly Dealers
Towing Hook System parts	Direct Fly Dealers
Electric Trim	Direct Fly Dealers or Ray Allen comp.
Strobe Lights	Direct Fly Dealers or Kunzelman
	SimplePack / Direct Fly Dealers
Landing Lights	Direct Fly Dealers
Cowling – Upper Section	Direct Fly Dealers
Cowling – Lower Section	Direct Fly Dealers
Avionics / Instruments	
VHF	Icom Dealers / Direct Fly Dealers
Transponder	Garmin Dealers / Direct Fly Dealers
GPS	Garmin Dealers / Direct Fly Dealers
ASI	Direct Fly Dealers or aircrafts part shop
Altimeter	Direct Fly Dealers or aircrafts part shop
Climb	Direct Fly Dealers or aircrafts part shop
Compass	Direct Fly Dealers or aircrafts part shop
Slip Indicator	Direct Fly Dealers or aircrafts part shop
Tachometer	Direct Fly Dealers or aircrafts part shop
Hourmeter	Direct Fly Dealers or aircrafts part shop
Oil Pressure	Direct Fly Dealers or aircrafts part shop
Oil Temperature	Direct Fly Dealers or aircrafts part shop
CHT	Direct Fly Dealers or aircrafts part shop
Fuel Level Indicator	Direct Fly Dealers
Flap Controller	Direct Fly Dealers
Circuit Breaker	Direct Fly Dealers or E.T.A company
Master/Start Key	Direct Fly Dealers

1.9. LIST OF DISPOSABLE REPLACEMENT PARTS

The following table shows a list of commonly replaced parts with its source.

ITEM	BRAND	REFERENCE # / SPECIFICATION	SOURCE
Engine and accessories			
Oil Filter	Rotax	825012	Rotax Dealers / Direct Fly Dealers / Aircraft Parts Stores
Air Filter	K&N	825551 / SP-2704	Rotax Dealers / Direct Fly Dealers / Aircraft Parts Stores
Fuel Filters	Direct Fly	-	Direct Fly Dealers
Engine Oil	SHELL	AeroShell Oil Sport PLUS 4	Shell Dealers / Direct Fly Dealers / Aircraft Parts Stores
Coolant	BASF CASTROL	Glysantin Protect Plus/G48 Castrol Radicool NF	BASF Dealers / Castrol Dealers/ Direct Fly Dealers / Aircraft Parts Stores
Tires, Brakes and Landing Gear			
Tires	Sava, SHENG SHIN	4.00-6/6 PR B11 , 13x5.00-6 B19	DF dealer/ Aircraft Parts Stores
Tubes	Sava, SHENG SHIN	4.00-6 TR 13 , 13x5.00-6	DF dealer/ Aircraft Parts Stores
Brake parts	Aerospool		DF dealer
Rubber shock absorber	Direct fly		DF dealer
Brake Fluid	Castrol	Castrol Brake Fluid DOT 4	Castrol Dealers / Direct Fly Dealers / Aircraft Parts Stores

1.10. ENGINE SPECIFICATIONS

ALTO aircraft is powered by ROTAX 912ULS engine rated at 100 hp.

Engine Type:	Normally-aspirated, liquid/air-cooled, dry sump, mechanical reduction drive, dual carburetor-equipped, four-cylinder, four-stroke, electronic dual ignition, horizontally-opposed cylinders with 73.91 cu.in. (1,2 l) displacement.
Maximum take-off power	73.5 / 100 kW / hp
Maximum continuous power.....	69.0 / 93 kW / hp
Maximum (5 min)	5,800 rpm
Maximum for continuous operation	5,500 rpm
Maximum cylinder head temperature	135 / 275 °C / °F
Maximum oil temperature	130 / 265 °C / °F
Minimum oil pressure (below 3,500 rpm)	12 psi
Minimum oil pressure (above 3,500 rpm).....	29 psi
Maximum oil pressure — cold engine start only.....	102 psi
Oil pressure — normal operation	29 – 73 psi
Fuel pressure — normal operation (min-max)	2 – 6 psi
Outside air temperature range.....	-25 to +50 / -13 to +122 °C / °F
Reduction Gear Ratio	2.43:1



Figure 1-3 – Rotax Engine

For more detail refer to the Rotax engine manual supplied with your engine.

1.11. WEIGHT AND BALANCE INFORMATION

Refer to the Pilot's Operating Handbook (DFE-POH-AR1), Section 6 for the Weight and Balance Data.

1.11.1. Definition

The removal or addition of equipment may result in changes to the center of gravity and empty weight of the aircraft. The permissible useful load can also be affected. In such case a new weight and balance is necessary to determine the new empty weight and center-of-gravity position. The new empty weight and C.G. position should be recorded in the Pilot's Operating Handbook, Section 6. Then a new permitted crew weight for fueling and baggage must be computed and recorded.

WARNING!

NEVER EXCEED THE MAXIMUM TAKEOFF WEIGHT AND C.G. RANGE FOR ANY CONFIGURATION OF CREW, FUEL AND BAGGAGE AS SHOWN IN THE AIRCRAFT OPERATING INSTRUCTIONS.

1.11.2. Empty weight determination

- Tools needed: Scales, ramp boards and rests under wheels to level the aircraft.
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:

The empty weight of an aircraft includes all operating equipment that has a fixed location and is actually installed in the airplane. It includes the weight of the painted airplane, accumulator, standard and optional equipment, full engine coolant, hydraulic fluid, brake fluid, oil and unusable fuel (2.0 liters / 0.5 US Gal). The aircraft is weighed without crew and baggage.

The following weighing procedure is recommended:

1. Remove excessive dirt, grease, moisture from the airplane before weighing
 2. Weigh the airplane inside a closed building to prevent errors due to wind
 3. Place the scales, calibrate zero
 4. Place the airplane on the scales (use boards to run on the scales or lift the airplane - see airplane jacking)
 5. Place the airplane in a level flight position (use suitable rests under the wheels)
 6. Check the configuration for weighing (e.g. empty weight);
 7. Weigh the airplane and record the values in Weight and Balance Record (make a copy of standard Record included in POH).
 8. Compute the weight and C.G. position according to the formula Weight and Balance Record
 9. Compute and record permitted crew weight for fueling and baggage - see POH, Section 6.
 10. Up-date the Weight and Balance information table and attach it in the List of the Supplements (Paragraph 9.2.1) of the POH.
- Task proper accomplishment check: Push a wing down to rock the aircraft and then repeat weighting. You should get the same results as before.

1.12. TIRE INFLATION PRESSURE

Tire pressures are noted on placards located on the airplane Landin Gears.

Nose wheel Tire	26 ± 3 psi	(1.8 ± 0.2 Bar)
Main Tire	26 ± 3 psi	(1.8 ± 0.2 Bar)

/For TUNDRA tires is pressure aprox 1.0 Bar - depend on the type of TUNDRA tires/

1.13. APPROVED OILS AND CAPACITIES

Refer to Paragraph 1.6.2.

1.14. RECOMMENDED FASTENER TORQUE VALUES

Metric thread		Strength class				
		4.8	5.8	8.8	10.9	12.9
M4	N.m kg.m		1,67 0,17			
M5	N.m kg.m		3,45 0,35			
M6	N.m kg.m	5,39 0,55	6,86 0,70	9,80 1,00	13,72 1,40	16,67 1,70
M7	N.m kg.m	8,82 0,90	10,78 1,10	14,70 1,50	20,59 2,10	25,49 2,60
M8	N.m kg.m	12,74 1,30	15,69 1,60	22,55 2,30	32,36 3,30	38,24 3,90
M10	N.m kg.m	24,51 2,50	31,38 3,20	44,12 4,50	61,78 6,30	73,54 7,50
M12	N.m kg.m	42,16 4,30	52,95 5,40	74,53 7,60	104,93 10,70	125,52 12,80
M14	N.m kg.m	66,68 6,80	78,54 8,00	117,67 12,00	164,75 16,80	196,13 20,00
M16	N.m kg.m	93,16 9,50	107,87 11,50	164,75 16,80	225,55 23,00	274,58 28,00
M18	N.m kg.m	137,29 14,00	171,61 17,50	245,16 25,00	343,23 35,00	411,87 42,00
M20	N.m kg.m	176,51 18,00	225,55 23,00	313,81 32,00	441,29 45,00	539,36 55,00
M22	N.m kg.m	225,55 23,00	284,39 29,00	392,26 40,00	558,97 57,00	676,65 69,00
M24	N.m kg.m	313,81 32,00	392,26 40,00	549,17 56,00	755,11 77,00	970,85 99,00
Ultimate strength (MPa)		420	500	880	1040	1220
ϕ in %		(14)	7	12	8	8
Yield point (MPa)		330	400	640	940	1100

Conversion: 1 N.m = 0.74 lb.ft

Torque moment formula (valid for all bolt sizes):

$$M_{kmax} = 1,065 \times \frac{d \cdot \sigma \cdot S}{1000 \cdot m}$$

Legend:

Mk.....torque moment N.m
d.....bolt shank diam. mm
σ.....min. yield point MPa
m.....safety factor (m=1,25 for σ<500 MPa; m=1,43 for σ>500 MPa)
s.....lead of helix mm

1.15. GENERAL SAFETY INFORMATION

There are many hazards inherently present when performing any maintenance task on this aircraft. To minimize the risk to both yourself and others, begin by thinking through each task that is to be performed before starting any work. Using common sense, think of ways to avoid these hazards. Remember also that many accidents happen because of carelessness. Be sure to also use the right tool for the task at hand and to use the proper personal protective equipment. Such equipment may include, but is not limited to:

- Eye protection – safety glasses, goggles, and face shield
- Gloves
- Hearing protection – ear plugs or muffs
- Apron
- Protective footwear with non-slip soles

You should also keep on hand a suitable fire extinguisher, absorbent material to contain spills, an eyewash bottle, and a general-purpose first aid kit. It is also advisable to have on hand the material safety data sheet (MSDS) for all products and chemicals that will be used during the servicing of the aircraft.

Some other general rules to follow are:

- Never leave the ignition switch or the master switch turned on when the engine is not running. Doing so could allow the engine to fire if the propeller were rotated by hand.
- Never operate the engine with untrained personnel around – everyone who is essential to be in the area should be reminded that a spinning propeller may be lethal yet almost invisible. Remove hats when running the engine to keep them from being removed by the propeller slipstream.
- Remove any loose clothing, such as neckties and scarves. Tuck in your shirt and secure any long hair to prevent them from becoming tangled in power tools.
- Remove all jewelry. Not only can items such as rings, watches, and necklaces become caught in rotating tools, they can also conduct electricity and may cause a short circuit. This could result in burns or damage to electrical circuits.
- Disconnect the negative lead from the battery when doing any electrical work that does not involve troubleshooting the electrical systems. This will reduce the risk of a short circuit or even a fire.
- The lead acid battery will emit hydrogen gas when charging, which is highly flammable. Any nearby source of ignition such as sparks or an open flame can result in an explosion. Keep all ignition sources away from the battery.
- Aviation gasoline is also highly flammable. When working with the fuel system, always work in a well-ventilated environment. Any nearby source of ignition such as sparks or an open flame can result in a fire or explosion. Keep all ignition sources away. Always ground the airframe to a suitable earth ground during fueling/defueling operations to reduce the risk of a static discharge ignition source.
- When working with the landing gear, always support the aircraft properly with jacks. Do not work underneath the aircraft unless it is properly supported.

1.16. FLIGHT SAFETY REPORTING INSTRUCTIONS

The owner/operator of a LSA is responsible for notifying the manufacturer of any safety of flight issue or significant service difficulty upon discovery.

Please report any service difficulties or any other issue relating to flight safety directly to Direct Fly using the form available on Paragraph 0.9. of this Manual. The form may be sent by email or fax to the addresses listed on Paragraph 0.1 on this Section

Section 2 – INSPECTIONS

2.1. GENERAL

This section is intended to serve as a guide for a person performing the maintenance tasks (certified airframe and powerplant mechanic, owner, LSRMA) to perform routine maintenance on the aircraft. It is the responsibility of the owner and/or the operator to maintain the aircraft in an airworthy condition and ensure that all applicable Safety Directives, Safety Alerts, and Service Bulletins have been complied with. Furthermore, it is the responsibility of the owner and/or the operator to ensure that the airplane is inspected as specified in Parts 43 and 91 of the Federal Aviation Regulations. This inspection guide is not intended to replace the good judgment of a certified airframe and powerplant mechanic.

The guide will make reference to service information provided by other vendors, such as the manufacturer of the engine. The persons performing the maintenance on the aircraft must ensure that they have the latest editions of these publications. This guide will not make reference to revision levels of vendor publications.

This guide will be applicable to the aircraft in the configuration it left Direct Fly S.R.O. when it was first delivered and it may not cover modifications made to the aircraft subsequently.

Inspection Groups and Criteria

Visual Inspection

Visual inspections will normally apply to those areas, surfaces, and/or items that become visible by the removal or opening of access doors, panels, fairings, or cowlings. Visual Inspection criteria will normally consist of, but are not limited to the following criteria:

- **Moving Parts**
Proper operation, correct alignment, security, sealing, cleanliness, lubrication, adjustment, tension, travel, condition, binding, excessive wear, cracking, corrosion, deformation, and any other apparent damage.
- **Metal Parts**
Security, condition of finish, cleanliness, distortion, fatigue cracks, cracked welds, corrosion, and any other apparent damage.
- **Fuel and Hydraulic Oil Lines and Hoses**
Cracks, dents, kinks, loss of flexibility, deterioration, obstruction, chaffing, improper bend radius, cleanliness, security, and any other apparent damage.
- **Electrical Wiring**
Cleanliness, loose, corroded, or broken terminals, chaffed, broken, or worn insulation; security; heat deterioration, and any other apparent damage.
- **Bolts and Nuts**
Fretting, wear, damage, stretch, proper torque and safety wiring.
- **Filters and Screens**
Filters and screens shall be removed, cleaned, inspected for contamination, or replaced as applicable.
- **Fuel Tank Areas**
Evidence of leaks, corrosion.

Operational Inspection

An operational inspection is a check intended to determine that a component or system is fulfilling its intended purpose. The operational inspection does not require quantitative tolerance.

Functional Inspection

When called for by an inspection task, a functional inspection is a quantitative check to determine if one or more functions of a component perform within specified limits. The functional inspection is a comparative examination of a component or system against a specific standard.

2.2. CONDITION INSPECTION TASKS

If the aircraft is registered in the United States or other country following the same or similar regulation, ASTM F2483 requires that all LSA category airplanes must undergo a complete inspection at least once every 12 calendar months. An authorized maintenance person, as described in ASTM F2483, must perform this inspection.

A signed and dated record must be maintained as each inspection task is completed. When the last task of the inspection has been completed, the Inspection Report is to be signed off in the log book/maintenance record. The inspection items to be covered in the condition inspection are identical to the 100-hour Inspection items. The inspection interval to the next condition inspection may not exceed twelve calendar months.

2.3. PERIODIC INSPECTION TASKS

If the aircraft is operated commercially (for hire) in the United States or other country following the same or similar regulation, it must also have an inspection every 100 flight hours. The 100-hour interval between inspections should never be exceeded by more than 10 hours, and then only if additional time is required to reach a place where the inspection can be satisfactorily accomplished. Additionally, the time the interval was exceeded must be included as flight hours in the next 100-hour interval. Inspection tolerances cannot be accumulated.

2.4. ACCESS WINDOWS

The following are the inspection and access windows:

- Screw caps on the wings lower surface - access to the aileron control rods and levers and to the pitot/static installation in the left half of the wing
- Screw cap on the fuselage lower surface under the baggage compartment close to the fuel tank - access to the fuel tank installation
- Screw cap on the fuselage lower surface in the middle of the rear section - access to the elevator control rods and a lever
- Wing fillets which cover space between the fuselage and wing - access to the wing-fuselage.
- Cover boot of Control stick system in the cockpit

2.5. SCHEDULE OF INSPECTIONS

- Pre-flight Inspection
- Post-flight Inspection
- After the first 25 ± 2 flight hours
- After every 50 ± 3 flight hours
- After every 100 ± 5 flight hours or 12 months (whichever comes earlier)

2.5.1. PRE-FLIGHT INSPECTION

- Tools needed: Screwdriver to remove upper engine cowling
- Parts needed: None
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions: Refer to Pilot's Operating Handbook section 4, Paragraph 4.1
- Task proper accomplishment check: None

A pre-flight inspection is performed prior to the beginning of each flight. A pre-flight inspection should be repeated prior to each flight even during the same day.

The Pre-flight inspection is a visual check of the aircraft for deformations, surface damage, fuel and oil system leaks, prop damage, released locks, covers and cowlings etc.

Any damage or failure should be repaired immediately if the airworthiness is affected or when the aircraft can not be put out of operation.

It is important to perform a pre-flight inspection carefully to prevent problems from arising.

Refer to the Pilots Operating Handbook for more details.

2.5.2. POST-FLIGHT INSPECTION

- Tools needed: Screwdriver to remove upper engine cowling
- Parts needed: None
- Type of maintenance: Preventive maintenance
- Level of Certification needed: Sport pilot or higher
- Instructions: Refer to Pilot's Operating Handbook section 4, Paragraph 4.1
- Task proper accomplishment check: None

Post-flight inspection is performed at the end of each flight day; the post-flight inspection events are the same as the preflight ones. If possible failures, damages and malfunctions should be recorded and repaired immediately. It is recommended to clean and/or wash the airplane and check that the fuel and oil consumption are in the normal range.

Lastly record all hours flown and other data in appropriate documentation (Log Book etc.).

2.5.3. PERIODICAL INSPECTIONS

- Tools needed: Set of basic metric tools, no specials tools required
- Parts needed: Refer to Paragraph 2.4.3.1
- Type of maintenance: Line
- Level of Certification needed: Refer to Paragraph 0.12.
- Instructions: Refer to Paragraph 2.4.3.1
- Task proper accomplishment check: Refer to Paragraph 2.4.3.1

The periods for overall checks and contingent maintenance will depend on the conditions of the operation and the overall condition of the airplane. The manufacturer recommends maintenance checks and periodic inspections in the following periods:

- 1) after the first 25 $\pm$ 2 flight hours
- 2) after every 50 $\pm$ 3 flight hours
- 3) after every 100 $\pm$ 5 hours of service or 1 times per year

Refer to the Rotax 912 Operator's Manual for engine maintenance.

Refer to the Propeller Maintenance Manual for propeller maintenance.

2.5.3.1. Periodical Inspection Actions / Checklists

The following Periodical Inspection Checklists are intended for copying and serve as the Maintenance Records. It is also recommended to include small repairs, damages and their remedy or replacement.

Some parts of the airplane (engine, propeller etc.) may have special time limits - refer to the appropriate manuals for maintenance time limits and procedures.

Model:	S/N.:	Hours flown:	Date of inspection:
Alto	Registration:	No. of Takeoffs:	

Event #	Event description	Inspection ☑ Tick off performed inspection				Carried out by:	Inspected by:
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
1.	Prior to the inspection clean and wash the airplane surfaces, if needed.	☐	☐	☐	☐		
2.	ENGINE	see engine manufacturer's instructions					
3.	ENGINE COMPARTMENT						
3.1.	Fiberglass engine cowlings						
3.1.1.	Check condition of cowlings and quick closing locks - repair any damage				☐		
3.1.2.	Remove engine cowling	☐	☐		☐		
3.1.3.	Visually check inside of cowling for wear marks or other defects, Visually check condition, attachment and security of air filter assembly		☐		☐		
3.2.	Engine mount						
3.2.1.	Visually check condition, attachment, security of attachment bolts: engine-engine mounting, engine mounting-firewall	☐	☐		☐		
3.2.2.	Visually check condition of rubber isolators - replace those cracked and excessively deformed				☐		
3.3.	Engine induction system						
3.3.1.	Check operation and condition of carburetor heat and induction air box	☐	☐		☐		
3.3.2.	Visually check condition of induction tubing	☐	☐		☐		
3.3.3.	Check carburetor - condition, control cables attachment, lubricate cables and cable attachment fittings at carburetor arms, inspect carburetor drain pans for condition and security	☐	☐		☐		
3.4.	Battery						
3.4.1.	Visually check attachment and security		☐		☐		
3.4.2.	Check charging - charge if needed				☐		
3.4.3.	Visually check condition and attachment of wire leads - replace those damaged	☐	☐		☐		
3.4.4.	Load test				☐		
3.5.	Wiring						
3.5.1.	Visually check condition and integrity of wires, connections, security of wires and grounding	☐	☐		☐		
3.6.	Fuel system						
3.6.1.	Visually check condition, integrity, attachment and security of hoses - replace those damaged	☐	☐		☐		
3.6.2.	Visually check fuel filter condition, clean filter element and inside of bowl	☐	☐		☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
3.6.3.	Visually check system for leaks	☐		☐	☐		
3.7.	Cooling system						
3.7.1.	Visually check radiator for condition and leaks				☐		
3.7.2.	Visually check condition, attachment of hoses; radiator left hose clearance from exhaust pipe approx. 0.8 in (20 mm), check system for leaks	☐	☐	☐	☐		
3.7.3.	Tighten hose clamps if needed		☐		☐		
3.7.4.	Check coolant quantity in the expansion tank - add or change coolant acc. to the engine manual if needed	☐	☐	☐	☐		
3.7.5.	Visually check condition and attachment of overflow bottle on the firewall and service to appropriate level				☐		
3.8.	Lubrication system						
3.8.1.	Visually check condition and attachment of oil tank				☐		
3.8.2.	Check oil cooler for condition, attachment and leaks	☐	☐	☐	☐		
3.8.3.	Visually check hoses for condition, leaks, attachment and security - replace damaged hoses	☐	☐		☐		
3.8.4.	Check oil quantity - add or change oil acc. to the engine manual if needed	☐	☐	☐	☐		
3.9.	Exhaust system						
3.9.1.	Visually check exhaust system for condition, cracks, deformations or damage - repair / replace. Check left front pipe clearance from radiator hose - approx. 0.8 in (20 mm),	☐	☐	☐	☐		
3.9.2.	Visually check condition and attachment of the muffler - repair / replace as needed	☐	☐	☐	☐		
3.9.3.	Check joint security	☐	☐	☐	☐		
3.10.	Heating						
3.10.1.	Visually check hoses leading air into the cockpit - check hose for condition, integrity, attachment and security		☐		☐		
3.10.2.	Check condition, function and control of the heating / venting system valves		☐		☐		
3.11.	Reinstall lower engine cowling Reinstall lower engine cowling when the inspection is completed and engine test run performed	☐	☐	☐	☐		
3.12.	Lubricate per Lubricating Chart	☐	☐		☐		
4.	PROPELLER	see manufacturer instructions +					
4.1.	Blades						
4.1.1.	Inspect blades for abrasions, cracks, paint damage, condition of blades leading edges and tips - repair according to the propeller manual	☐	☐	☐	☐		
4.2.	Spinner						
4.2.1.	Visually check spinner for condition, abrasions, cracks, paint damage - repair as necessary		☐	☐	☐		
4.2.2.	Remove spinner		☐	☐	☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
4.3.	Propeller	see manufacturer instructions +					
4.3.1.	Check prop attachment, security of bolts		☐		☐		
4.3.2.	Check run-out				☐		
4.3.3.	Check all assembly and mounting bolts for proper torque per manufacturers manual.		☐	☐	☐		
4.3.4.	Install spinner				☐		
4.3.5.	Intentionally left blank						
5.	LANDING GEAR						
	NOSEWHEEL LANDING GEAR						
5.1.	Nose wheel leg						
5.1.1.	Check condition and attachment of the nose wheel leg (lift airplane nose)	☐	☐	☐	☐		
5.2.	Wheel pants						
5.2.1.	Visually check wheel pants or mudguards condition - repair damages and cracks		☐		☐		
5.2.2.	Remove fairing (reinstall when nose wheel inspection is completed)				☐		
5.3.	shock cord and rubber suspension stop						
5.3.1.	Visually check shock cord and suspension stop and flexible washer for deformation, cracks, excessive wear - replace if needed		☐		☐		
5.4.	Tire						
5.4.1.	Check tires for condition, cuts, uneven or excessive wear and slippage - replace if needed		☐		☐		
5.4.2.	Check tire pressure - inflate to required pressure	☐	☐	☐	☐		
5.5.	Wheel						
5.5.1.	Visually check for cracks, permanent deformations - if damaged, replace				☐		
5.5.2.	Check valve condition around the hole in the rim				☐		
5.5.3.	Check condition of bearings, wheel free rotation, play				☐		
5.6.	Joints						
5.6.1.	Check torque and security of fixed joints	☐	☐		☐		
5.6.2.	Check nose wheel free rotation inside the leg - the rotation should not be too free to prevent shimmy		☐		☐		
5.7.	Nose wheel control system						
5.7.1.	Check control rods condition, rod ends security		☐		☐		
5.7.2.	Check condition of nose wheel control lever covers for wear through - repair damage				☐		
5.8.	Lubricate per Lubricating Chart	☐	☐		☐		
6.	LANDING GEAR						
	MAIN LANDING GEAR						
6.1.	Fiberglass legs						
6.1.1.	Visually check condition of fiberglass legs - repaint damaged areas, contact airplane manufacturer if cracks were found	☐	☐	☐	☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
6.1.2.	Inspect leg attachment into the fuselage (no play) - Lift the landing gear and move the leg forward-backward, upward-downward; at the same time check wheel play on the axle - tighten attachment bolts if the leg has a play		☐	☐	☐		
6.1.3.	Check torque and security of fixed joints	☐	☐		☐		
6.1.4.	Check cloth cover which covers the leg-fuselage input hole				☐		
6.1.5.	Check grounding				☐		
6.2.	Wheel pants or mudguards						
6.2.1.	Visually check wheel pants / mudguards condition - repair damage and cracks		☐		☐		
6.3.	Tires						
6.3.1.	Check tires for condition, cuts, uneven or excessive wear and slippage - replace if needed	☐	☐		☐		
6.4.	Wheel						
6.4.1.	Visually check wheel rims for cracks, permanent deformations - replace wheel rim in case of cracks				☐		
6.4.2.	Check valve condition around the hole in the disc				☐		
6.4.3.	Check condition of bearings, wheel free rotation, play		☐		☐		
6.5.	Brakes						
6.5.1.	Check attachment of brake system plastic hoses to the main leg				☐		
6.5.2.	Visually check condition of pads - steady and symmetry abrasion of pads - replace pads if needed		☐		☐		
6.5.3.	Check wear of the disc and overheating				☐		
6.5.4.	Check brake system for leaks - add brake fluid and bleed the system if a brake pedal has soft movement	☐	☐	☐	☐		
6.5.5.	Brake fluid replacement, clean up and reseal brake cylinders as needed						
7.	WING						
7.1.	Wing						
7.1.1.	Visually check condition - no loose rivets, deformations, cracks or any other damage - repair small damage or contact the airplane manufacturer	☐	☐		☐		
7.1.2.	Check play of wing suspensions - move the wing tip upward-downward, frontward-rearward				☐		
7.1.3.	Check condition and attachment of fiberglass wing tips				☐		
7.1.4.	Visual check of riveted joints leakage of integral tank				☐		
7.1.5.	Check fuel sump drains for leaks, drain sumps	☐	☐	☐	☐		
7.1.6.	Check wire integrity and grounding				☐		
7.1.7.	Check lines				☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
7.2.	Aileron						
7.2.1.	Visually check condition	☐	☐		☐		
7.2.2.	Check free movement	☐	☐		☐		
7.2.3.	Check aileron hinge	☐	☐		☐		
7.2.4.	Check play		☐		☐		
7.2.5.	Check security of control rod ends		☐	☐	☐		
7.2.6.	Check grounding		☐		☐		
7.2.7.	Remove inspection covers from the lower wing surface to inspect security of control system joints		☐		☐		
7.2.8.	Lubricate per Lubricating Chart and reinstall covers.	☐	☐		☐		
7.3.	Flap						
7.3.1.	Fully extend the flaps and visually check condition	☐	☐		☐		
7.3.2.	Check flap hinge	☐	☐		☐		
7.3.3.	Check play		☐		☐		
7.3.4.	Check condition of flap control pin and wear of the groove at the flap root		☐		☐		
7.3.5.	Lubricate per Lubricating Chart	☐	☐		☐		
7.4.	Pitot-static tube						
7.4.1.	Check pitot-static tube attachment				☐		
7.4.2.	Check pitot-static system for leaks - the airplane manufacturer uses KPU 3 instrument				☐		
7.4.3.	Check hoses under pitot-static tube (in bend)				☐		
7.4.4.	Check fall sensor of AOA system if installed.				☐		
7.4.5.	Check safety system of AOA system if installed.						
7.5.	Wing suspensions						
7.5.1.	Remove wing fillets	☐	☐		☐		
7.5.2.	Visually check condition of wing suspensions,	☐	☐		☐		
7.5.3.	Check wear, corrosion				☐		
7.5.4.	Check security of joints	☐	☐		☐		
7.6.	Lubricate per Lubricating Chart	☐	☐		☐		
8.	FUSELAGE						
8.1.	Fuselage surface						
8.1.1.	Visually check condition - no loose rivets, deformations, cracks or any other damage, corrosion - repair small damage or contact the airplane manufacturer	☐	☐		☐		
8.1.2.	Visually check rivets near the landing gear attachment				☐		
8.1.3.	Check condition and attachment of equipment - antenna, beacon etc. (Inspect ELT for condition and function per FAR 91.207 (d) Verify ME 406 registration date and battery expiration date.)				☐		
8.1.4.	Check tail skid attachment				☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
8.1.5.	Visually check condition, attachment and operation of towing mechanism (if installed) and lubricate		☐	☐	☐		
8.1.6.	Visually check condition of fiberglass wing fillets				☐		
8.2.	Cockpit canopy						
8.2.1.	Visually check canopy condition for - cracks, scratches, any other damage - drill end of cracks(ref. maintenance manual for limits or contact manufacturer)	☐	☐		☐		
8.2.2.	Check canopy lock for condition and operation	☐	☐		☐		
8.2.3.	Check vents for condition and operation				☐		
8.2.4.	Inspect safety and alignment pins on forward canopy framework and rail stops of canopy for condition and security				☐		
8.2.5.	Check canopy silicone rubber packing				☐		
8.2.6.	Check cabin lighting				☐		
8.2.7.	Check cabin dewatering				☐		
8.2.8.	Check placards/labels				☐		
9.	HORIZONTAL TAIL UNIT						
9.1.	Visually check condition - For loose rivets, deformation, cracks, deep scratches and any other damage, ref. maintenance manual or contact the manufacturer for instructions	☐	☐		☐		
9.2.	Visually check condition and attachment of fiberglass tips				☐		
9.3.	Check elevator free movement	☐	☐		☐		
9.4.	Check elevator hinge	☐	☐		☐		
9.5.	Check play - move the stabilizer frontward-rearward, upward-downward - contact the airplane manufacturer if play exceeded tolerances of maintenance manual		☐		☐		
9.6.	Check security of joints at control sticks	☐	☐		☐		
9.7.	Aileron and Elevator Trim tab						
9.7.1.	Visually check condition		☐	☐	☐		
9.7.2.	Check hinge		☐	☐	☐		
9.7.3.	Check electrical trim - functionality				☐		
9.7.4.	Check securing the linkage adjusting screws				☐		
9.7.5.	Check trim indicators for proper function				☐		
9.8.	Lubricate per Lubricating Chart	☐	☐		☐		
10.	VERTICAL TAIL UNIT						
10.1.	Visually check condition - For loose rivets, deformation, cracks, deep scratches and/or other damage ref. maintenance manual or contact the manufacturer for instructions	☐	☐		☐		
10.2.	Visually check condition and attachment of fiberglass tips				☐		
10.3.	Check rudder free movement	☐	☐		☐		
10.4.	Check rudder suspensions	☐	☐		☐		
10.5.	Check play - move rudder upward-downward	☐	☐		☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☒ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
10.6.	Check joints security	☐	☐		☐		
10.7.	Lubricate per Lubricating Chart	☐	☐		☐		
11.	COCKPIT						
11.1.	Instrument panel						
11.1.1.	Visually check condition and attachment of the instrument panel		☐		☐		
11.1.2.	Check condition and attachment of individual instruments		☐		☐		
11.1.3.	Check function of instruments				☐		
11.1.4.	Check throttle and choke levers free movement and lock	☐	☐		☐		
11.1.5.	Inspect completeness and readability of placards				☐		
11.1.6.	Check wire integrity				☐		
11.1.7.	Check heating controls and parking brake				☐		
11.2.	Seats						
11.2.1.	Visually check seat upholstery, remove upholstery				☐		
11.2.2.	Visually check seats and backrests condition				☐		
11.2.3.	Check for loose rivets or any other damage on the seats pan areas				☐		
11.2.4.	Visually check main landing gear legs attachment inside the fuselage				☐		
11.3.	Safety harness						
11.3.1.	Visually check condition, attachment and security				☐		
11.4.	Hand control						
11.4.1.	Remove aileron rod covers inside the cockpit		☐		☐		
11.4.2.	Check hand control free movement	☐	☐	☐	☐		
11.4.3.	Check play	☐	☐		☐		
11.4.4.	Check joints security	☐	☐		☐		
11.4.5.	Check stops for condition				☐		
11.4.6.	fuel lines for condition				☐		
11.4.7.	Lubricate per Lubricating Chart	☐	☐		☐		
11.5.	Rudder control						
11.5.1.	Check stiffness of movement	☐	☐	☐	☐		
11.5.2.	Check joints security	☐	☐		☐		
11.5.3.	Check stops on firewall and nose gear				☐		
11.5.4.	Check condition , security and proper tension of cables	☐	☐		☐		
11.5.5.	Check hydraulic brake system for leaks, attachment of master cylinders and fluid level of resivor on outside of firewall - add brake fluid if needed	☐	☐	☐	☐		
11.5.6.	Lubricate per Lubricating Chart	☐	☐		☐		
11.6.	Flap and trim control, Towing mechanism control						
11.6.1.	Check free movement of levers	☐	☐	☐	☐		
11.6.2.	Check operation of flap control automatic and manual system operation , function of lock of manual override switch and indicator lights for proper indication.	☐	☐		☐		
11.6.3.	Lubricate per Lubricating Chart	☐	☐		☐		

Event #	Event description	Inspection				Carried out by:	Inspected by:
		☑ Tick off performed inspection					
		after the first 25 hrs.	after the first 50 hrs.	every 50 hrs.	every 100 hrs.		
11.7.	Complete lubricating per Lubricating Chart	☐	☐		☐		
11.8.	Install seats upholstery and covers						
11.9.	Engine Test Run (see FM) * idling * throttle and choke levers operation * acceleration - deceleration * r.p.m. drop with either magneto switched off * max. r.p.m. * test brake system efficiency	☐	☐	☐	☐		
11.10.	Test flight as required for level of service, check compliance with service directives for aircraft and installed components, check all required documentation is on board aircraft.	☐	☐	☐	☐		

Section 3 – STRUCTURES

3.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft primary structures by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft structures.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

3.2. FUSELAGE

3.2.1. Description

The fuselage is a combination of semi monologue structure consisting of reinforcements and duralumin skin. Fuselage section is rectangular in the lower part and ecliptic in the upper part. The fin is an integral part of fuselage. The two place cockpit is located in the middle part of the fuselage that is accessible after forward opening sliding canopy. The engine compartment in the front part of the fuselage is separated from the cockpit by the stainless steel firewall to which the engine bed is attached.

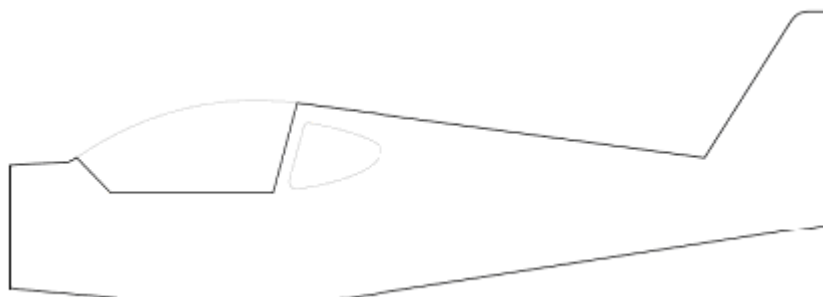


Fig 3-1: Fuselage

Cockpit

The cockpit has a side-by-side, dual control arrangement. The instrument panel is located in front of the crew. The flap control lever is located on the vertical quadrant between seats. Elevator trim is electric, with the control button located on the central console between the seats.

A baggage compartment is situated behind the seats with maximum load of 33 lb (15 kg)

The cockpit floor is covered with a removable carpet and the seats are also covered with cloth or leather upholstery. The interior cockpit sides are covered with upholstered panels containing pockets.



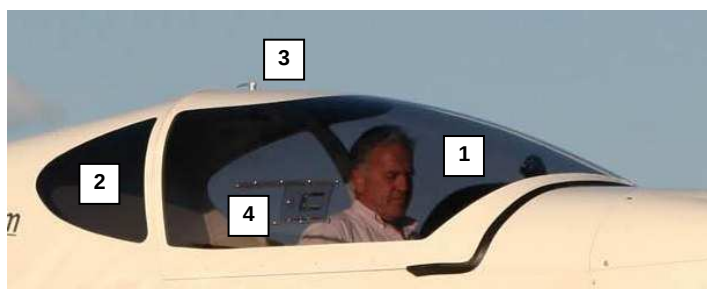
Figure 3-2 – Cockpit Layout

- **Cockpit controls**

Refer to the PILOT'S OPERATING HANDBOOK – POH- Section 7 - for more details about the Cockpit Controls for the specific aircraft.

- **Cockpit canopy**

The plexi-glass (perspex) canopy consists of a single forward sliding assembly, attached to a aluminum tube frame. Rear side windows are made of plexi-glass (perspex). The canopy is attached to the fuselage by two sliding rails allowing it to slide forward. For ease of maintenance, the canopy can be removed completely by depressing two locking pins on the forward section of the canopy rail. The canopy is equipped with a latch on the upper rear section of the frame.



- | | |
|----|------------------------|
| 1- | forward sliding canopy |
| 2- | rear window |
| 3- | canopy lock |
| 4- | ventilation window |

Figure 3-3 –Cockpit Canopy Layout

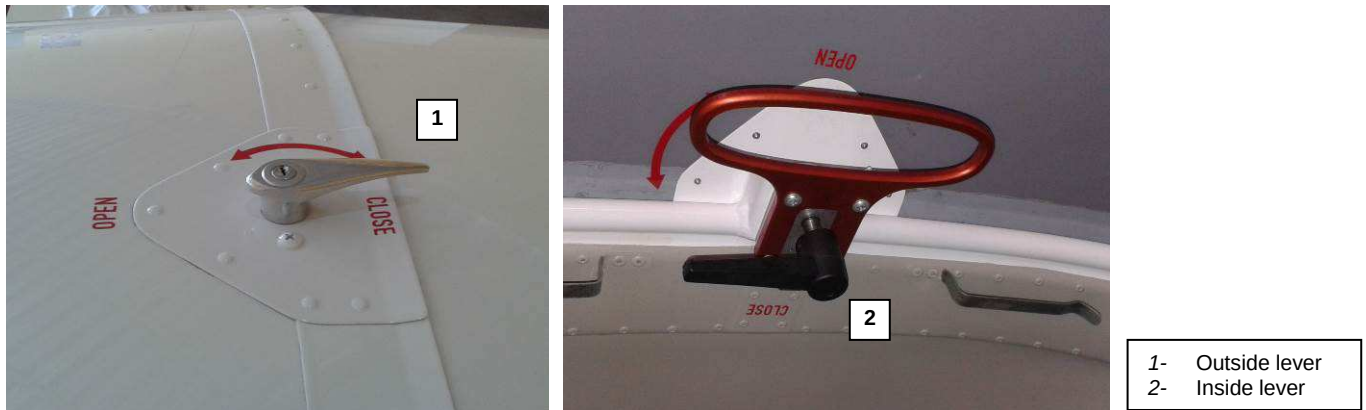


Figure 3-4 & 3-5 – Cockpit canopy lock

CAUTION

Keep the canopy locked during taxiing, flying and engine run test.

- **Seats and safety harness**

The plane has two side-by-side seats, each equipped with four-point seatbelts. The waist belt is attached alongside the seat and shoulder belts behind the seats.

- **Baggage compartment**

The baggage compartment is situated behind the seats. It is equipped with the net (optional) to prevent baggage movement. Maximum baggage weight is stated on a placard located near the compartment.



Figure 3-6 – Safety Seat Belts and Baggage Compartment

3.2.2. Instructions for Authorized Maintenance, Repair or Alterations

3.2.2.1. Sliding Cockpit Canopy Maintenance

- Tools needed: None
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed: Sport pilot or higher, Aircraft owner (only for ELSA)
- Instructions:

Canopy Removal: (2 people required)

1. Slide the canopy forward
2. One person will press the black plastic lock located on the inner side of the canopy frame and the second will press the black plastic lock located on the inner side of the other side of the canopy. Both people have to press the plastic lock at the same time which will allow the canopy to slide from the fuselage rails.



Fig 3-7 to 3-10: Canopy Disassembly sequence

Canopy Mounting: (2 people)

1. Slide the two fuselage rails as forward as possible.
 2. Two people will carry the canopy and will slide the canopy onto the fuselage rails. Move the canopy forward and the black plastic locks will lock the canopy to the sliding rails.
- Task proper accomplishment check: Check proper attachment of the canopy, open and close it. Check if it may be locked in closed position.

3.3. WING

3.3.1. Description

The wing is of rectangular shape, consisting of reinforcements and duralumin skin, single-spar structure with the auxiliary spar with suspended ailerons and wing flaps. Riveting is used for connecting individual structural elements. Fiber-glass wing tips are riveted on the wing ends. There is an integral fuel tank inside each wing in front of the main spar in the leading edge of the wing.

Ailerons

The ailerons are of rectangular shape on each half of the wing and are attached to the wing with hinges. An aileron is formed with the ribs and cover, which forms a hollow section.

Trim tab

The right wing aileron is equipped with an adjustable trim tab, Optionally it can be fitted with an electrically controlled trim tab.

Flaps

Two-third's of each wing is fitted with a flap. The flaps are of rectangular shape and are formed with the ribs and cover, which forms a hollow section. The flap is attached to the wing with a hinge (3 hinges).

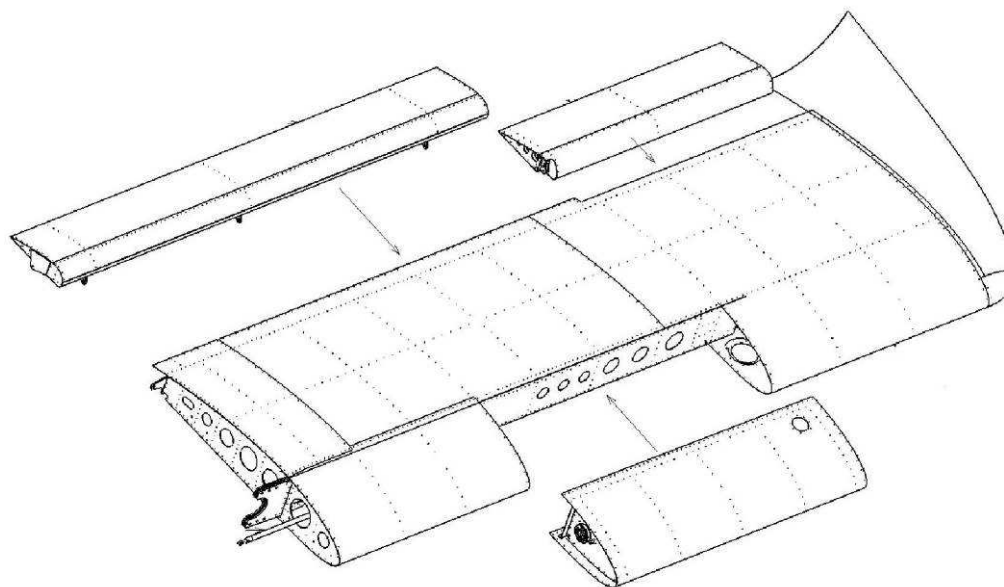


Fig 3-11: Wing

3.3.2. Instructions for Authorized Maintenance, Repair or Alterations

3.3.2.1. Wing to Fuselage Assembly

The wing assembly procedure for an aircraft is the following: (3 persons are needed to accomplish this task):

- Tools needed: Light Hammer, Metric Wrenches, Torque Meter, Screwdriver
- Parts needed: Bolts, nuts, washers and grease.
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Thoroughly clean and lubricate all the wing suspensions and bolts with a suitable lubricant before the assembly. Also lubricate the flap root groove.
 2. The first person holds the wing tip, the second person holds the wing root leading edge, while the third holds the wing root trailing edge.

CAUTION

Take care of the pitot-static tube(s) when handling the wings.

3. Set the wing carefully on the wing attachments on the fuselage in such a way that the wing flap is set with the slot on the control pin. When sliding the wings on the attachments take increased care so that damage to hoses of pitot-static system (left half of the wing) and electric system cables cannot occur.
4. Set the wing so that the attachments on the wing and on the fuselage are concentric.
5. The person keeping the wing on the leading edge will insert the bolts into the upper main attachment (the bolt head is in flight direction). Shift bolt by means of slight hammering to the stop (shifting can be facilitated by slight moving the wing tip up and down). Thereafter insert the bolt into the lower main attachment and shift it by slight hammering to the stop. Insert other 4 bolt into the holes.
6. Insert the bolt into the rear attachment of the wing (the bolt head is in flight direction) and push it by slight hammering to the stop. Put the washer on the bolt and screw the nut on it.
7. Put on the washers on the wing main attachment bolts and screw the nuts on them. Apply torque as per table 1.14 on Section 1.
8. Connect the aileron control pull rod, secure the joint.
9. Connect fuel hoses through the holes in the fuselage and join the fuel hoses to the opposite parts and secure them with hose clip.
10. Connect wiring.
11. Install pitot-static system hoses and carry out leakage test of the pitot-static system.
12. Install wing fairings.

- Task proper accomplishment check:
 - Visually check that all bolts are inserted and secured properly.
 - Visually check if the Aileron control pull rod are connected and secure.
 - Visually check if fuel hoses are connected and if the clips are installed
 - Visually check if wiring is connected
 - Visually check if pitot-static hoses are connected.

3.3.2.2. Wing from Fuselage Disassembly

The wing disassembly procedure for an aircraft is the following: (3 persons are needed to accomplish this task):

- Tools needed: Light Hammer, Round Timber, Metric Wrenches, Torque Meter, Screwdriver
- Parts needed: None
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Remove the fuselage-wing fairings.
 2. Drain all fuel from tanks. Push airplane tail down (almost to the ground) to allow drainage of the unusable fuel.
 3. Prior to removal of the left wing disconnect the pitot-static hoses.
 4. Disconnect fuel hoses from beaks on the fuselage
 5. Disconnect cable plugs and sockets of electrical system.
 6. Disconnect aileron control pull rod.
 7. The first person will hold onto the wing tip, the second person by the root of the leading edge, the third person by the root of the trailing edge.
 8. Push out the safety bolt securing the crown nut of the rear bolt and dismantle the rear bolt of the wing attachment.
 9. Release bolts on the main attachments of the wings.
 10. Relieve the wing by slight lifting the wing tip upwards.
 11. Use a hammer and round timber to knock out the lower and the upper bolt from the main wing attachments.
 12. Pull the wing away from the fuselage to disconnect the wing from the fuselage.
 13. Position the disconnected wing in such a way that it won't be damaged.
- Task proper accomplishment check: Visually check that nothing was damaged on the airplane due to wing removal.

3.4. EMPENNAGE

3.4.1. Description

- **Horizontal tail unit (HTU)**

The HTU of conventional type consists of the stabilizer and elevator with the trim tab. Single-spar structure of the HTU consists of duralumin ribs, spar and skin. The top view of HTU is of rectangular shape.

Stabilizer

The stabilizer is rectangular in shape and formed with a “duralumin” cover and ribs. The stabilizer is attached to the fuselage with two bolts at the leading edge and secured with two bolts at the stabilizer trailing edge.

Elevator

The elevator is rectangular in shape and formed with a duralumin cover and ribs. The elevator is attached to the stabilizer with a hinge. There is also a hinged trim tab on the left elevator.

Trim tab

The left elevator is equipped with a servo controlled trim tab of rectangular shape. The tab is formed with “duralumin” plate.

- **Vertical tail unit (VTU)**

The VTU consists of the fin and rudder. The rudder is attached to the fin by two hinges. The frame of the VTU is composed of a metal sheet spar and a duralumin cover.

Fin

The fin is attached to the fuselage rear section and is formed with a duralumin spar and cover. The fin tip is formed with a fiberglass cap.

Rudder

The rudder is formed with a duralumin spar and cover and attached by two hinges at the fin. The rudder upper tip is formed with a fiberglass cover..

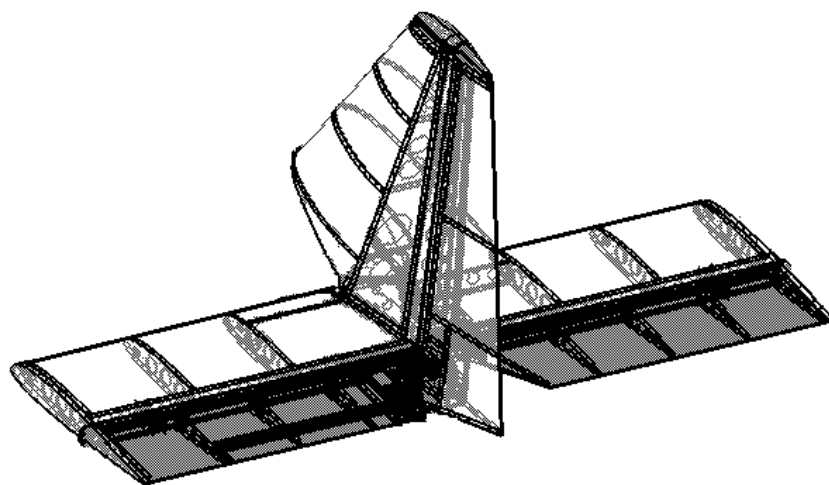


Fig 3-12: Empennage

3.4.2. Instructions for Authorized Maintenance, Repair or Alterations

3.4.2.1. HTU to Fuselage Assembly

- Tools needed: Metric Wrench, Torque Meter.
- Parts needed: Bolts, nuts, washers and grease.
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Make the connecting components ready, clean and lubricate HTU suspensions.
 2. Insert the HTU from the rear into the fuselage as far as the stabilizer will go into the two pins in the front, and the two bolts (M8) in the rear
 3. Put the washers on the M8 bolts. Screw on the nuts
 4. Insert the M8 bolt to connect the elevator control hinge with the control rod. Put on a washer, and self-locking nut.
 5. Connect trim tab wires.
 6. Adjust the elevator and trim tab deflections (see 3.6.2.2)
- Task proper accomplishment check:
 - Check proper attachment of trim control wires
 - Check HTU proper attachment and securing of all joints
 - Check elevator and trim tab deflection

3.4.2.2. HTU From Fuselage Disassembly

- Tools needed: Metric Wrench.
- Parts needed: None.
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Unscrew attachment bolts of HTU-fuselage fiberglass cover.

1. Disconnect the trim tab wires.
 2. Disconnect the elevator control cables.
 3. Remove the safety pins securing the castle nuts on the bolts of the stabilizer rear suspensions. Screw off the nuts and remove the washers.
 4. Draw the HTU out of the fuselage.
 5. Put connecting components in a safe place to avoid losing them
- Task proper accomplishment check: Visually check that nothing was damaged on the airplane due to HTU removal.

3.4.2.3. Rudder Assembly

- Tools needed: Metric Wrench
- Parts needed: Bolts, nuts, washers and safety pins.
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Put the rudder on the fin suspensions from above. Use care not to move the spherical bearings in the rudder suspensions.
 2. If necessary insert a washer to adjust lower suspension clearance.
 3. Put the washer on the lower suspension bolt, tighten the castle nut and secure with a safety pin.
 4. Attach the rudder control cables. Tighten the castle nut and secure with safety pins.
- Task proper accomplishment check: Check rudder proper attachment and securing of all joints.

3.4.2.3. Rudder Disassembly

- Tools needed: Metric Wrench
- Parts needed: None.
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Disconnect the rudder control cables; attach the ends of the cables together to keep the cables from slipping inside the fuselage.
 2. Remove the safety pin from the lower suspension bolt. Remove the castle nut and washer.
 3. Lift and remove the rudder from suspensions.
- Task proper accomplishment check: Visually check that nothing was damaged on the airplane due to rudder removal.

3.5. LANDING GEAR

3.5.1. Description

- **Main landing gear**

The main landing gear consists of the left and right landing gear legs. The legs are formed from composite / fiberglass and are fixed by means of bolts in the fuselage casing under the seats. Wheel axles are bolted to the lower part of the main landing gear legs. The main wheels on both legs are equipped with hydraulic disc brakes controlled by a hand lever in the central console between the seats. The wheels can be covered with the fiberglass fairings (wheel pants).

- **Nose wheel landing gear**

The steerable nose landing gear consists of a front landing gear leg, a rubber bungee suspension unit and a suspension stop. The nose leg is made of a steel tube, attached to the firewall by two bearings. The axle, with wheel attached, is connected to the welded bushing at the bottom part of the leg. Two rods are used for the leg steering by the control pedals.

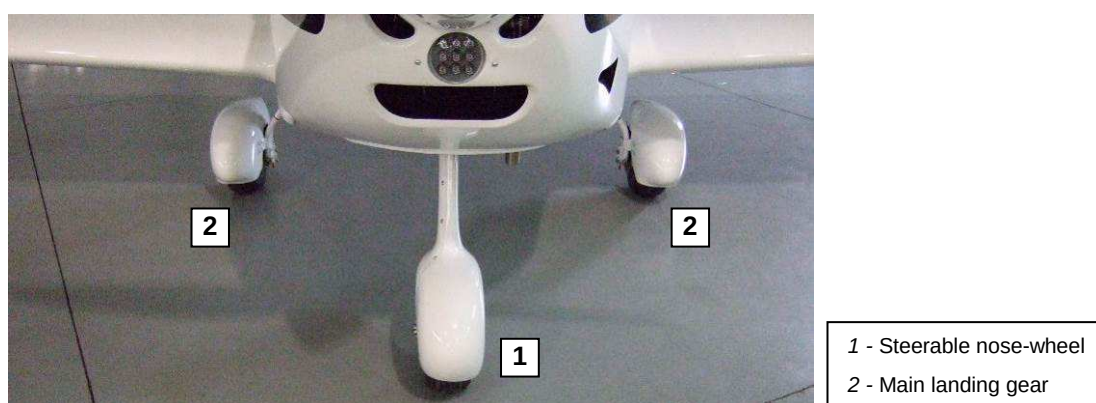


Figure 3-13 – Landing Gear

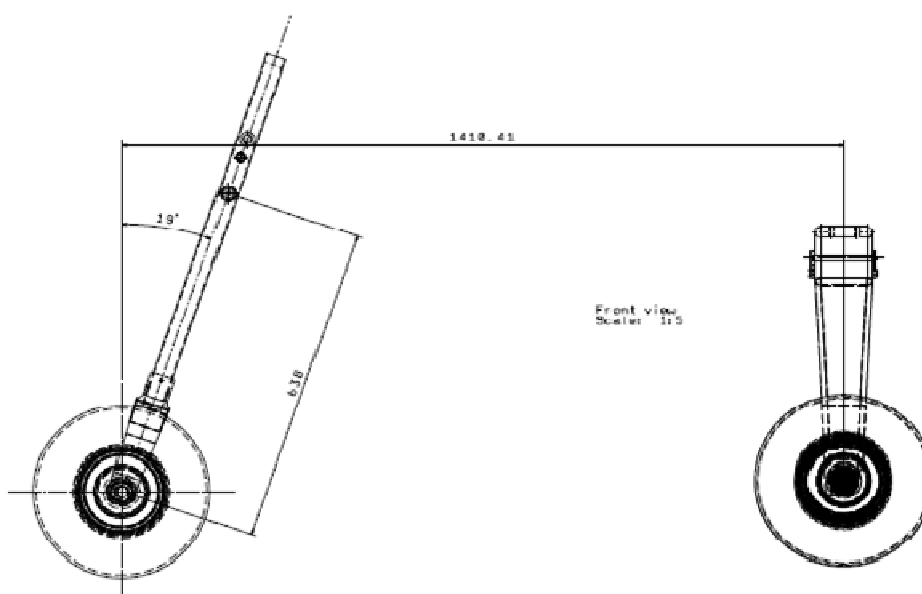


Figure 3-14 – Steerable nose wheel landing gear layout

- **Auxiliary tail skid**

The auxiliary tail skid is attached at the lower rear part of the fuselage and protects the aircraft from inadvertent damage during tail-down landing conditions. The tail skid is formed from composite/fiberglass.



Figure 3-15 & 3-16 – Auxiliary tail skid

- **Wheel brakes**

Both wheels on the main landing gear are equipped with hydraulic disc brakes. The brake system consists of the hydraulic master cylinder, plastic hoses, calipers with cylinder, brake pads and the brake disc which is bolted onto the inner part of the rim.



- | |
|--|
| 1- Brake caliper with the hydraulic cylinder |
| 2 - Brake disc |
| 3 - Brake fluid hose |
| 4 - Terminal |
| 5 - Air bleed screw |

Figure 3-17 - The Aerospool brake on the wheel

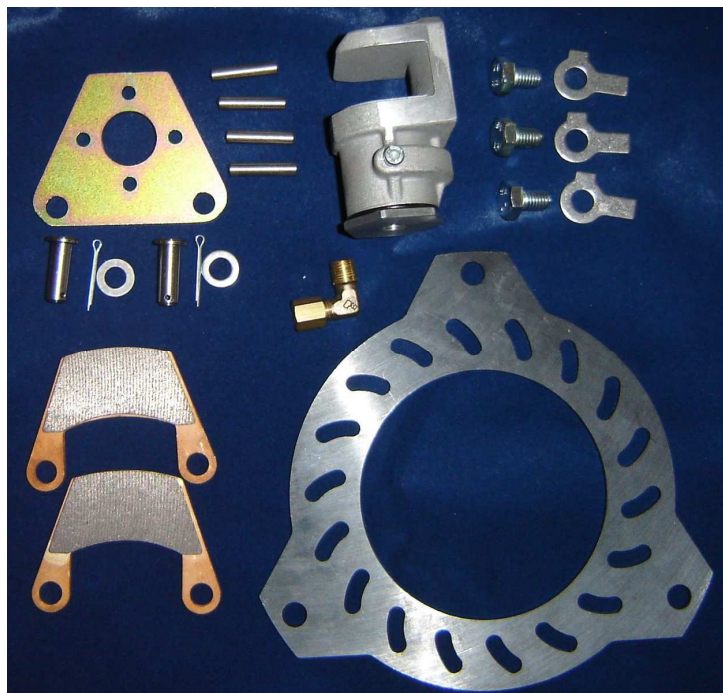
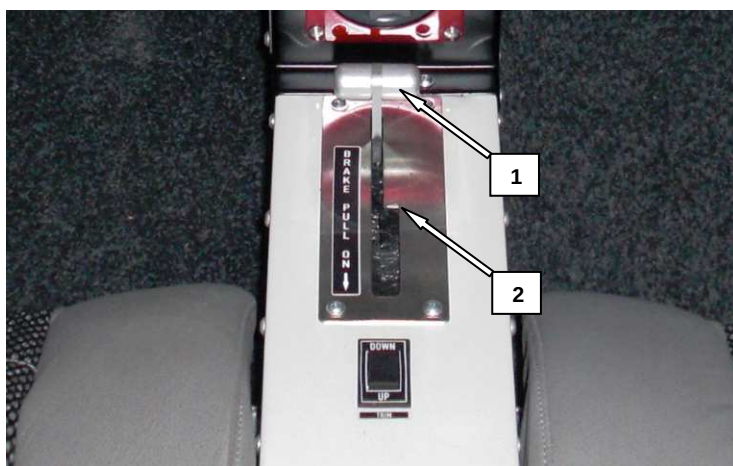


Figure 3-18 - The Aerospool brake

- **Brake control**

The brakes on both wheels are controlled by a hand lever mounted on the central console between the seats.

Parking Brake function is available by locking the lever on the right side recess on the plate.



- | | |
|----|------------------------------------|
| 1- | Brake Control Lever |
| 2- | Recess for the parking brake lock. |

Figure 3-19 – Brake Control

- **Wheels**

All the wheels consist of a two-part casting rim with a tire and tube. The main wheels are on an axle attached to the main gear leg, fastened by the nuts.

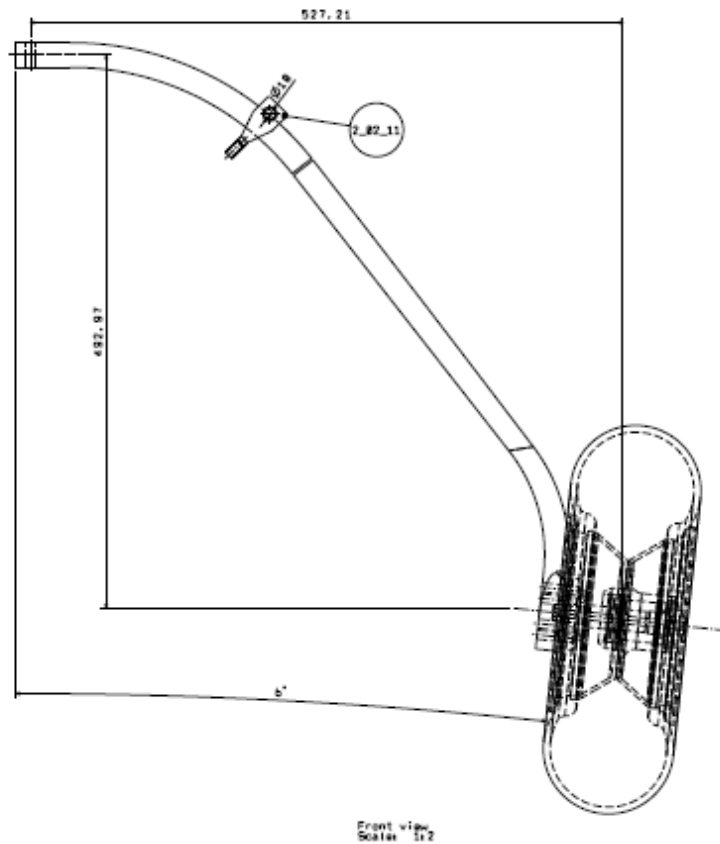


Figure 3-20 – Main Landing Gear Wheel Layout

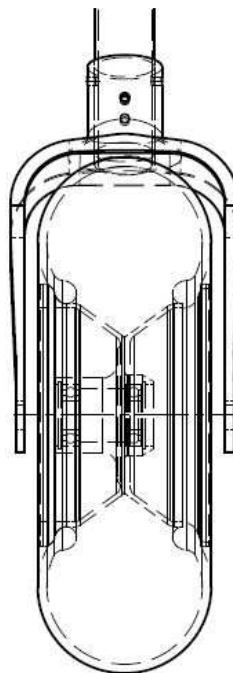


Figure 3-21 – Nose Landing Gear Wheel Layout

- **Tires and Tubes**

All Tires	main SAVA 14x4, 4.00-6/6 PR B11 front SHENG SHIN 13x5.00-6 B19
Tubes	main 4.00-6 TR 13 front 13x5.00-6
Main wheel rim.....	AEROSPOOL 02-A Veral
Nose wheel rim.....	AEROSPOOL 02-A Veral
Master Cylinders.....	MAGURA 012 2020 225.4

3.5.2. Instructions for Authorized Maintenance, Repair or Alterations

3.5.2.1. Disassembly of main wheel pants

- Tools needed: Metric Allen Wrenches
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:

Main wheel pants are attached with 8 bolts to a bracket mounted on the main landing gear leg and with one bolt from outside to the wheel axle.



Figure 3-22 – Wheel Pants Bracket

1. Remove 8 bolts attaching the wheel pant from inner side to the bracket.
 2. Remove the bolt attaching the wheel pant to the wheel axle.
 3. Remove Wheel Pant
- Task proper accomplishment check: None.

3.5.2.2. Assembly of main wheel pants

- Tools needed: Metric Allen Wrenches
- Parts needed: Bolts, washers, spacing tubes and locking wire
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher

- Instructions:

When assembling the main landing gear wheel pant, proceed as follows:

1. Set the wheel pant to position and bolt it with 4 bolts to the bracket.
 2. From outside insert the bolt with washer into the hole in the wheel pant. From inner side of the wheel pant, shift the spacing tube on the bolt and screw the bolt into nut hole in the wheel center. Secure the bolt head with locking wire in order to prevent from its turning.
- Task proper accomplishment check:
 - Visually check that all inside bolts are inserted and secured properly.
 - Visually check if the outside bolt is secured by the locking wire.

3.5.2.3. Disassembly of nose wheel pant

- Tools needed: Screwdriver, Metric Allen Wrench
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher

- Instructions:

Nose wheel pant is assembled with 2 bolts in a bracket to the nose leg, 22 bolts in its side cover and 2 bolts fixed in the center of the front wheel axle.



Figure 3-23 – Nose Wheel Pant

1. Unscrew the bolts from the wheel axle
2. Unscrew the bolts from the side cover and remove it.
3. Unscrew the bolts on the bracket connected to the nose leg
4. Remove the wheel pant.

- Task proper accomplishment check: None.

3.5.2.3. Assembly of nose wheel pant

- Tools needed: Screwdriver, Metric Allen Wrench
- Parts needed: Bolts and washers.
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:
 1. Assemble the bracket connected to the nose leg.
 2. Screw the bolt from the left side of the wheel axle
 3. Screw the bolts of the side cover
 4. Screw the bolt from the right side of the wheel axle
- Task proper accomplishment check: Visually check that all bolts are inserted and secured properly.

3.5.2.4. Disassembly of main landing gear wheel

- Tools needed: wrench to tighten/remove the M13 , M21 nuts Metric, Screwdriver
- Parts needed: new tube
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Support the aircraft to lift a wheel with punctured tire.
It is recommended to support the aircraft near the main gear leg entry into the fuselage (a reinforced area)
 2. Remove the wheel pant if installed
 3. Deflate tire
 4. Remove cotter pin securing castle nut from the axle
 5. Remove the castle nut and washer from the axle
 6. Cut the safety wire securing the bolts which join together the rim halves
 7. Remove the 3 bolts connecting the rims together
 8. Remove outer rim from the axle
 9. Remove tire
 10. Replace the tire and/or tube
 11. Insert a new tube into a tire and pump up slightly (put easy powder on the tube and/or in the tire)
- Task proper accomplishment check: none

3.5.2.5. Assembly of main landing gear wheel

- Tools needed: Metric wrench, Screwdriver
- Parts needed: split pins, self-locking mats
- Type of maintenance: Line
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
 -
- Instructions:
 1. Replace the tire and/or tube
 2. Insert a new tube into a tire and pump up slightly (put easy powder on the tube and/or in the tire)
 3. Put the valve insert of the tire with tube into the hole on the rim
 4. Put the rim with tube and tire onto the axle and adjust the wheel hub
 5. Insert the 6 connecting screws and slightly tighten
 6. Adjust the tire and tube positions to avoid puncturing the tube
 7. Tighten the screws
 8. Put the washer and distance tube on the axle, control the position of distance tube.
 9. Screw the castle nut on the axle
 10. Insert a new cotter pin and lock the castle nut
 11. Secure the screw heads with safety wire - pull the safety wire through the holes in two screw heads and twist together.
 12. Install the wheel pant

- Task proper accomplishment check: check tire pressure

3.5.2.6. Brake Pads Replacement

- Tools needed: Screwdriver, Metric Wrench
- Parts needed: brake pads , Loctite 638 to secure the screws,
- Type of maintenance: Line
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Jack the airplane
 2. Remove the cotter pin, unscrew the slotted nut M16, and remove the washer from the axle
 3. Bend small tabs on the 3 washers and unscrew the M6 screws connecting the brake disc to the wheel rim
 4. Remove the wheel and the distance ring from the axle
 5. Take the brake disc off (leave the brake on a main leg)
 6. Remove the cotter pins, shift the pins out and remove the brake pads
 7. Mount a new brake pad, secure the pins with cotter pins (diam. 2mm, length14mm) (part. No.: 039 300)
 8. Put the brake disc on the wheel
 9. Put the distance ring and the wheel on the axle (adjust the distance ring between bearings)
 10. Set tab washers (6x2, Nom. 038160) on the screws, apply Loctite and attach the brake disc to the inner part of the rim. Bend the tabs of the washers to secure the screw heads.
 11. Put the washer on the axle, tighten the slotted nut and secure with a cotter pin
- Task proper accomplishment check: check braking function, new brake pads require some running-in period to achieve maximum braking efficiency. This caution should be kept in mind considering safety of operation.

3.5.2.7. Bleeding

- Tools needed: Screwdriver, Metric Wrench set, plastic bottle for outgoing brake fluid,
- Parts needed: brake fluid
- Type of maintenance: Line
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Open the central panel C - according to 1.6.4
 2. Check maximum level of brake fluid
 3. Push back repeatedly on the braking lever and setup the lever in parking position
 4. Slowly open bleeding screw, outgoing brake fluid with air bubbles put down in the plastic bottle
 5. Tighten the screw
 6. Repeat several times: 2 - 5
 7. close the central panel, install parking T grip
- Task proper accomplishment check: check braking function

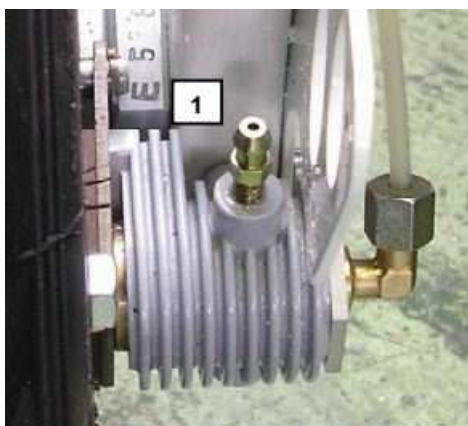


Figure 3-24 – Brake Bleeding Screw

NOTE

If the brake efficiency remains unsatisfactory after bleeding or if the pedal motion is excessive, fill with brake fluid and bleed the system again. Continue until all the air is out of the system.

3.6. STRUCTURAL CONTROL SURFACES

3.6.1. Description

- **Longitudinal control system**

The airplane is equipped with a classic dual control system. The elevator is controlled by control sticks, with connecting rods and arms. A control stick push/pull movement is transmitted, by a rod inside the quadrant between the seats to the elevator through a two-armed lever located underneath the baggage compartment floor cover.

From there two control cables with turnbuckles transmit a longitudinal motion towards the two-armed lever of the elevator. A control stick motion is limited by two stops. Both control sticks have a common "push-down" stop on the center-section and each control stick has a "pull-up" stop formed with a reinforcement riveted on the front edge of each seat. The rods have adjustable ends and swivel bearings to adjust the elevator deflections.

- **Lateral control system**

The ailerons are controlled by control sticks, with connecting rods and arms. A control stick lateral motion is transferred by a short rod in the cockpit to a longitudinal movement of a longer rod in the wing. This in turn transfers to the angular displacement of a two-armed lever attached to the wing main spar. The two-armed lever angular movement is transferred to the ailerons by short rods. The rods have adjustable ends to adjust the aileron deflections. The control stick has a termination stop.

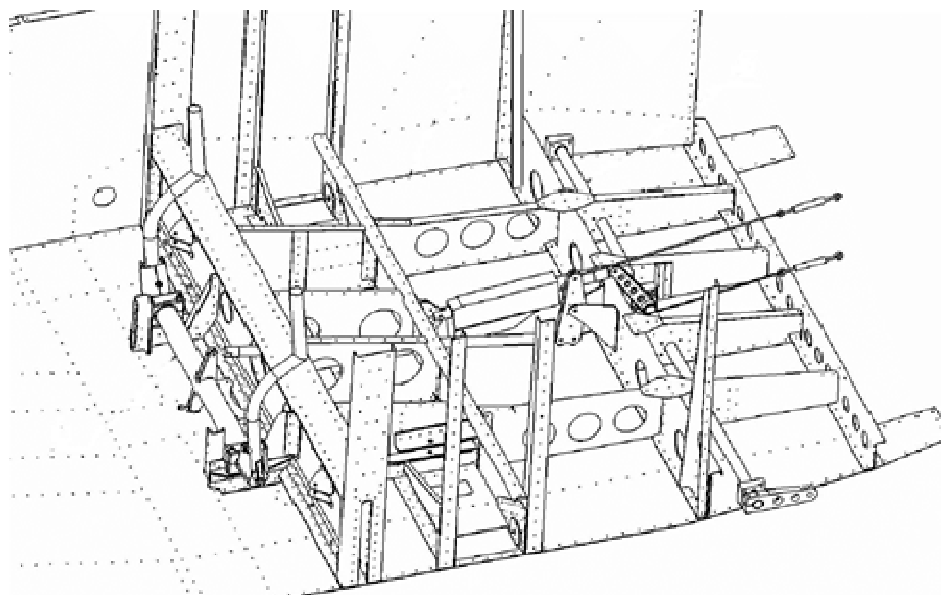


Figure 3-25 – Longitudinal and lateral control system layout (Fuselage)

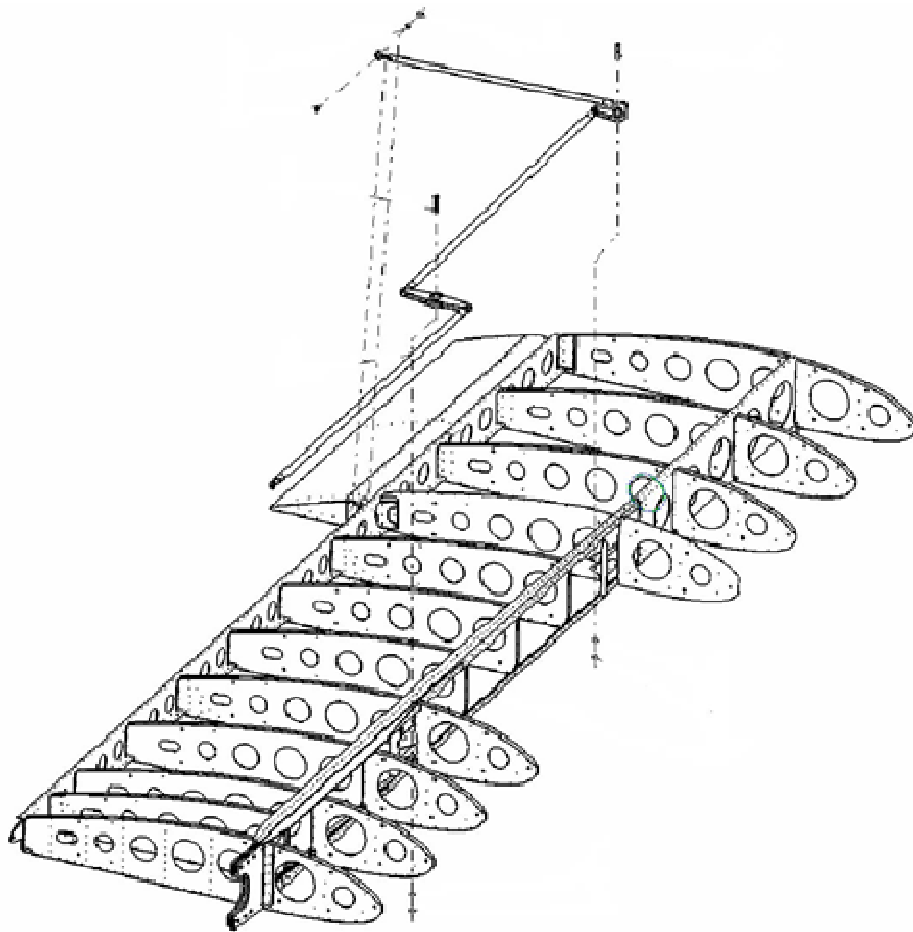
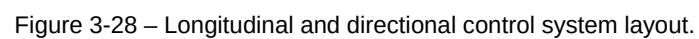


Figure 3-26 – Lateral control system layout (Wing)

- **Directional control system**

The rudder is controlled by cables attached to the rudder pedals and guided along the fuselage center to the rudder. The rudder pedals are attached to the cockpit floor. There are toe brake pedals on both the pilot and copilot side to operate the main wheel brakes. The cables are connected to the hinges in the lower part of the rudder spar. The cables are pre-stressed by means of nose wheel control rods. The rudder control is connected to the nose wheel landing gear to control the nose wheel by the adjustable rods. Rudder stops are located on the front part of the firewall bulkhead by "T" weldment on the nose wheel leg.



- **Flap control system description**

The wing flaps are controlled by an electronic controller located on the vertical quadrant below the instrument panel. The servo push/pull actuator movement is transferred to a longitudinal movement of a rod behind the seats (torsional tube). Then to an angular displacement of a lever onto a tube connecting left and right flap.

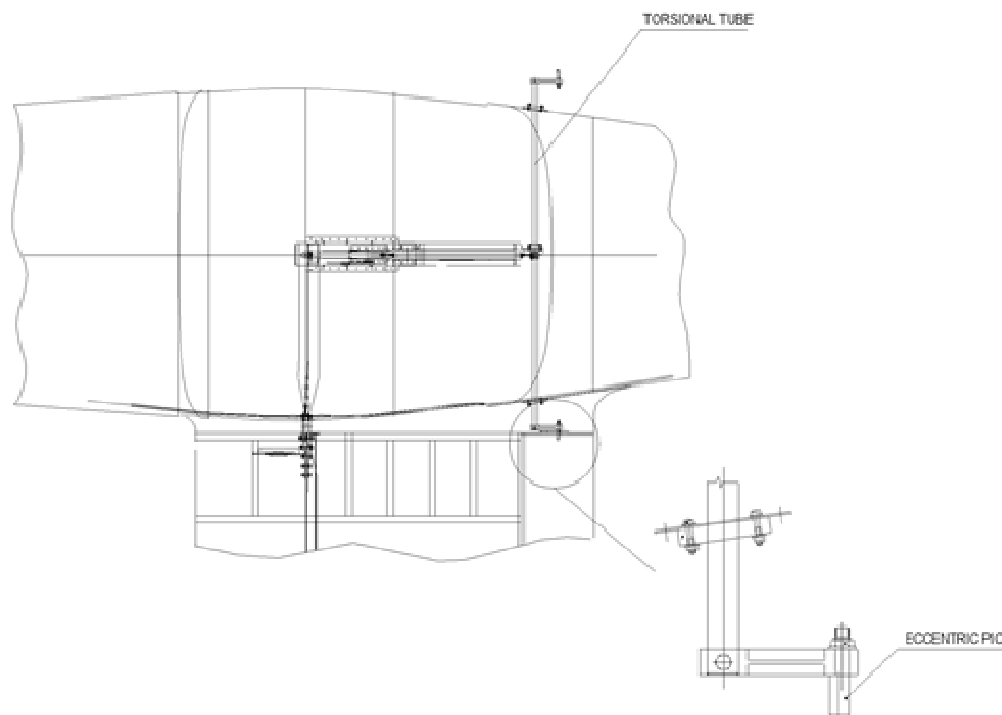


Figure 3-29 – Flap control system layout

- **Elevator Trim tab control system description**

A electric elevator trim control is located between the seats. Trim tab position indicator is displayed on the indicator on the panel.

A servo on the elevator is interconnected to the trim tab by means of a push rod and clevis arrangement.

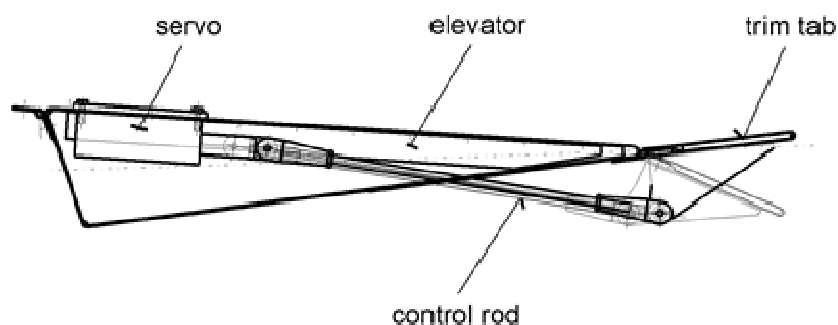


Figure 3-31 – Elevator trim tab control layout.

- Nose wheel control system description (optional)

The aircraft is equipped with steerable nose wheel landing gear. The wheel control system is connected to the rudder pedals. The nose wheel control system consists of the rods, connecting pedals and a two-armed lever welded to the nose landing gear leg.

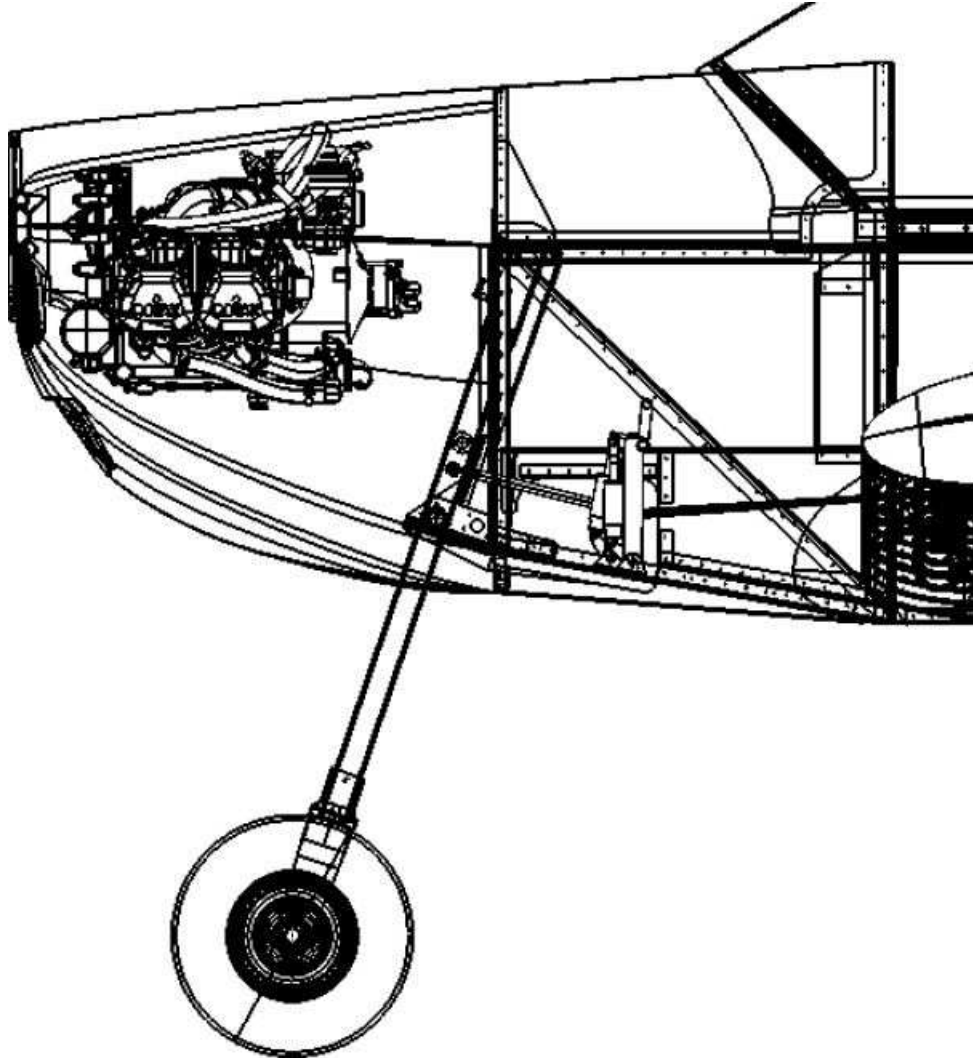


Figure 3-32 – Nose wheel control system layout.

3.6.2. Instructions for Authorized Maintenance, Repair or Alterations

3.6.2.1. Measurement of control surfaces deflections

- Tools needed: Protractor with a deflection hand, a clip to attach protractor to measured control surface.
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:

Required deflections

- Elevator	
Pos:	30°
Neg:	20°
- Aileron	
Pos:	28°
Neg:	25°
- Rudder	
Left:	30°
Right:	30°
- Trim Tab (Elevator)	
Pos:	15°
Neg:	30°
- Trim Tab (Aileron)	
Pos:	25°
Neg:	25°
- Flap:	
Position I	15°
Position II	25°
Position III	45°

Aileron deflection measurement

1. Attach a protractor with a deflection hand at the aileron upper surface by means of a clamp.
2. Set the aileron in neutral position (the aileron must fit the wing profile).
3. Zero the protractor - starting position for measurement.
4. Deflect the aileron fully down/up and note the deflections.
5. Compare, the measured deflections with the ones specified in the Control Surfaces Deflection Requirements. If required - adjust the aileron.

NOTE

If a protractor is not available, the following procedure may be substituted:

1. Insert a stiff cardboard sheet of paper in the space between the aileron and the flap. Hold the drawing paper against the flap.
2. Put the aileron in a neutral position. Then trace its profile (upper surface from the hinge to the trailing edge). Then trace the profile of the aileron fully deflected in both directions.
3. Remove the drawing paper and measure the deflection from the neutral position using a protractor.
4. Compare the measured deflections with those specified in the Control Surfaces Deflection Requirements. If required - adjust the aileron

Flap deflection measurement

1. Cut a strip of aluminum sheet 2 inches (50 mm) wide. The strip is then attached to the flap lower surface with two bolts (somewhere in the middle of the flap span where two nuts are riveted). The strip should overhang the flap trailing edge.
2. Attach a protractor with a deflection hand at the strip using a clamp.
3. Zero the protractor - this will be the starting position for measurement with the flaps retracted
4. Extend the flap to the required position using the flap control lever and read the deflection
5. Compare the measured deflections with those specified in the Control Surfaces Deflection Requirements. If required - adjust flap.

NOTE

If a protractor is not available, the following procedure may be substituted:

1. Insert a stiff cardboard sheet of paper in the space between the aileron and the flap. Hold the drawing paper against the flap.
2. Trace the profile of the retracted flap on the lower surface from the hinge to the flap trailing edge)
3. Move the flap to an extended position and trace the lower surface profile again
4. Remove the drawing paper and measure the deflection from the "RETRACTED" position using a protractor
5. Compare the measured deflections with those specified in the Control Surfaces Deflection Requirements. If required - adjust flap.

Elevator deflection measurement

The starting position to measure the elevator deflections is the neutral positions of the control stick and elevator. The neutral position of the control stick is set by the aircraft manufacturer, by means of a jig. The distances between the control stick and instrument panel, and between the control stick and fuselage side can be used to set the neutral position. When the elevator is in the neutral position, the chord of the Horizontal tail unit will be parallel to upper edge of the fuselage side (lower frame of the cockpit).

1. Attach a protractor with a deflection hand at the elevator trailing edge
2. Set the elevator to the neutral position
3. Zero the protractor
4. Fully pull or push the control stick to deflect the elevator and read the deflection
5. Compare the measured deflections with those specified in the Control Surfaces Deflection Requirements. If required - adjust elevator deflection.

NOTE

If a protractor is not available, the following procedure may be substituted:

1. Support the airplane under the tail skid and firewall and set the airplane in a horizontal position (a level set on the canopy lower frame can be used to set the airplane in horizontal position)
2. Stand a suitable staff close to the elevator trailing edge and mark the neutral position of the elevator.
3. Move the control stick and fully deflect the elevator. Mark the positions of the elevator while fully deflected
4. Measure the distances between marks on the staff
5. Compare the measured deflections with those specified in the Control Surfaces Deflection Requirements. If required - adjust elevator deflection.

Rudder deflection measurement

The rudder deflections are set by the aircraft manufacturer. A measuring instrument is used by the aircraft manufacturer to measure the rudder deflections. The instrument is put on the vertical tail unit and a rudder deflection may be read directly.

The rudder deflection may be measured however, when the set the rudder is set to the neutral position. Stand a suitable staff at the ruder trailing edge and mark lower edge of the rudder. Fully deflect the rudder and measure using a ruler between the mark on the staff and the lower edge of the rudder. Compare the measured distance with that specified in the Control Surfaces Deflection Requirements.

Trim Tab deflection measurement

The trim tab deflection is measured from the neutral position. When the trim tab profile does not protrude from the elevator profile with elevator set in neutral position.

1. Attach a protractor with a deflection hand at the trim tab
2. Neutralize the trim tab and the elevator
3. Zero the protractor
4. Set the trim tab in maximum lower or upper position using the trim tab control lever and read the deflection from the protractor scale.
5. Compare the deflection with that specified in the Control Surfaces Deflection Requirements. If required - adjust trim tab deflection.

NOTE

If a protractor is not available, the following procedure may be substituted:

1. Insert a stiff cardboard sheet of paper in the space between the elevator and the trim tab and hold the cardboard against the elevator.
2. Trace the profile of the neutralized trim tab
3. Move the trim tab to the maximum (both directions) using the trim tab control lever and trace the profile again
4. Remove the cardboard and measure the deflection from the neutral position using a protractor
5. Compare the deflection with that specified in the Control Surfaces Deflection Requirements. If required - adjust trim tab deflections.

- Task proper accomplishment check: None.

3.6.2.2. Control surfaces deflection setting

- Tools needed: Protractor with a deflection hand, a clip to attach protractor to measured control surface, Metric Wrenches, Screwdriver.
- Parts needed: 1.6 x 16 mm Cotter Pins
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Control surface deflections of a new aircraft are set by the manufacturer. Deflections are adjusted to values specified in the Control Surfaces Deflection Requirements as specified on Paragraph 3.6.2.1. of this Section.

A neutral position of the control surfaces and controls is used as a base for adjustment of deflections.

Aileron deflection adjustment

Control stops are on the stick

A range of deflections are set using adjustment screws on lateral control stops located on a console of the left/right control column.

The aileron neutral position can be adjusted with the adjustable end of the short rod inside the cockpit. Or with the adjustable end of the longer rod inside the wing (when the screw cap on the lower wing surface is removed).

The aileron differential (difference between a lower and upper deflection) can be adjusted with the adjustable end of the rod connecting the two-armed lever inside the wing and the hinge. The adjustable rod end is accessible when an aileron is deflected upward.

CAUTION

To adjust an aileron deflection, never use the adjustable end of the short rod which is accessible when wing fillet (covering the space between the wing and fuselage) is removed.

Flap deflection adjustment

Use the following procedure to adjust flap deflection:

Remove the baggage compartment bottom cover which will allow access to a rod connecting the flap control lever and a lever welded on a tube connecting the left and right flap. Use the adjustable rod end to adjust the flaps in the position "RETRACTED" or you can use eccentric pin to set retracted position – see figure 3-29.

The deflection of the left flap against the right flap may be adjusted when the eccentric pin is moved a slight amount. The pin is inserted into the groove of the flap root.

Elevator deflection adjustment

The range of elevator deflection is determined by the control stick stops setting. The "Push in" stop may be adjusted with an adjustment screw. To adjust an upward deflection, fully pull the control stick rearward. Then measure the elevator upward deflection and compare it with the one specified – if required adjust the deflection with the adjustable end of the elevator control rod or with the adjustable end of the rod below the baggage compartment cover. When the upward elevator deflection is adjusted, fully push the control stick and adjust a stop so that an elevator downward deflection corresponds with that one specified.

Rudder deflection adjustment

The rudder deflections are set by the aircraft manufacturer by fixed stops on the firewall contacting the weldment of the nose gear strut tube.

Trim Tab deflection adjustment

Trim tab deflection may be adjusted using the adjustable links attached to the trim actuators.

- Task proper accomplishment check:
 - check securing of all joints
 - check proper tightening of all screws
 - check the deflections comply with Control Surfaces Deflection Record

3.6.2.3. Steerable nose wheel landing gear adjustment

- Tools needed: Metric Wrenches
- Parts needed: None.
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

A steerable nose wheel adjustment is necessary so the airplane tracks during straight taxiing with engine idling (to eliminate prop turning moment) with rudder pedals held in neutral position (no crosswind!):

1. Lift the nose wheel and neutralize wheel and rudder pedals
 2. Adjust the nose wheel control using the adjustable rod ends.
- Task proper accomplishment check: Check straight taxiing at idle with rudder pedals neutralized

3.6.2.4. Rubber shock absorber replacement

- Tools needed: Metric Wrenches
- Parts needed: New rubber shock absorber, cotter pins.
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Perform rubber shock absorber replacement when rubber rope is damaged or excessively worn down.

1. Remove cotter pins from the hollow pin attaching a rubber rope
2. Remove hollow pin
3. Remove rubber rope from the nose landing gear leg
4. Mount a new rubber rope at the leg, sling over pin
5. Insert a hollow pin back
6. Secure by two cotter pins

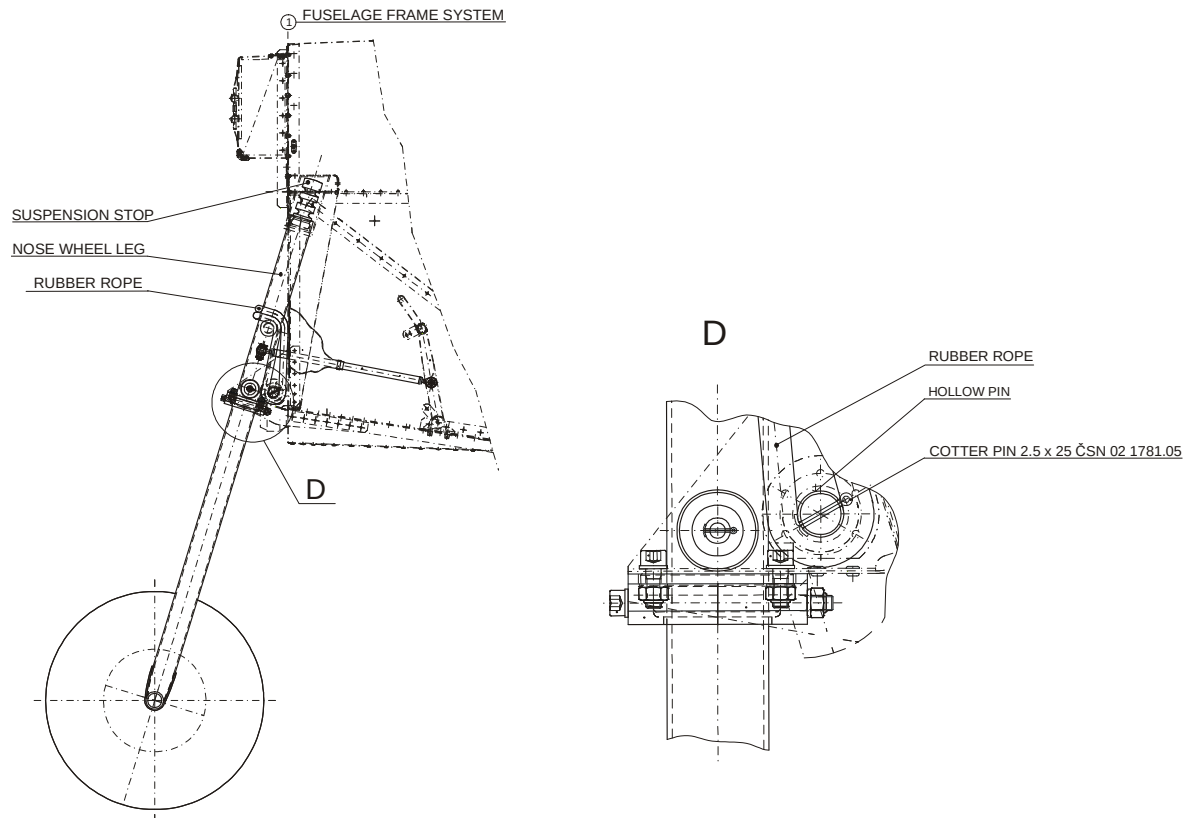


Figure 3-33 – Rubber shock absorber.

- Task proper accomplishment check:
 - Check function of nose wheel shock absorbing.
 - Check for proper installation of cotter pins.

3.7. PERMISSIBLE TOLERANCES

Description

The following table indicates the permissible control for critical control the airplane. These values should not be exceeded in operation.

Control expected that an operator will system steps if excessive play are found on/in part not listed below.

3.7.2. Instructions for Authorized Maintenance, Repair or Alterations

Clearance control and Excessive Play Correction

- Tools needed: Dial Indicator
- Parts needed: swivel bearing, pins if replacement is necessary to be consulted with aircraft manufacturer.
- Type of maintenance:
 - Preventive maintenance (Clearance control)
 - Line (Excessive Play Correction)
- Level of Certification needed: Preventive maintenance (Clearance control)
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Level of Certification needed: Line (Excessive Play Correction)
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

System	Instructions to find an excessive play	Instructions to correct an excessive play	Max. component play	Max. operational play
Ailerons control system	Block ailerons up to the wing and move the control to the left and right	Check condition of bearings and replace if needed	0.08 in 2 mm	0.2 in 5 mm
Control system	Block control up to the stabilizer, pull and push the control	Check condition of bearings and replace if needed	0.08 in 2 mm	0.2 in 5 mm
Flaps control system	Set the flaps in all control by control and then handle the flap trailing edge near the flap root, move the trailing edge up/downward to find possible plays	Check the part with oval hole for the control pin in the flap root rib and replace the worn-out pin or the part with oval hole.	0.08 in 2 mm	0.2 in 5 mm
Wing-Fuselage attachment	Move the wing tip and note play in wing suspensions (play is measured on the wing tip).	Check wing suspensions, replace pins	0.08 in 2 mm	0.16 in 4 mm
HTU attachment	Move the stabilizer tip control-rearward	Replace bearings in suspension points and bearings in control system. Check nuts tightening and add distance washers, respectively	0	0.08 in 2 mm
Rudder hinges	Lift the rudder	Change swivel bearing or insert a washer under the lower hinge pin	0.04 in 1 mm	0.08 in 2 mm
Nose wheel	Push the rear part of the fuselage down (use a weight) to lift the nose wheel, then move the wheel forward- rearward	Remove the wheel, remove the rim and tire and replace the bearings	0.04 in 1 mm	0.12 in 3 mm
Main landing gear	Lift the wing tip (hold the wing under the main spar) to lift a main leg, then move the wheel forward-rearward and note play in bearings or leg attachment	Check the leg attachment, wheels attachment, replace the bearings, if necessary	0.04 in 1 mm	0.12 in 3 mm

- Task proper accomplishment check: Verify if all clearances are within specified on the table after correction.

Section 4 – ENGINE

4.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft engine and correlated items by the Aircraft owner (only for ELSA), a Sport Pilot or higher, or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

Refer to the BRP-Rotax Maintenance Manual, supplied with the Aircraft documentation, for any other maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, that are not covered by this Maintenance Manual.

4.2. ENGINE

4.2.1. Description

The Rotax 912 ULS installed on Alto is a 4-stroke, 4 cylinder, horizontally opposed, spark ignition engine and has one central camshaft, push-rods, OHV, liquid cooled cylinder heads, ram air cooled cylinders, dry sump forced lubrication, and dual breaker-less capacitor discharge ignition.

The engine is fitted with an electric starter, AC generator and mechanical fuel pump. Prop drive is via reduction gear with integrated shock absorber.

For more information about the technical data of the engine, go to Paragraph 1.10, on Section 1 of this Manual.

Refer to the Rotax documentation for more details about different versions.



Figure 4-1 – Rotax Engine

Lubrication system description

The Rotax 912 engine is provided with a dry sump forced lubrication system. The oil pump pulls the engine oil from the oil tank attached to the firewall via the oil cooler and then forces it through the oil filter to the lubrication points in the engine.

The oil emerging from the lubrication points accumulates on the bottom of the crankcase and is forced back to the oil tank by the blow-by gases.

The oil tank is equipped with a vent hose.

The engine lubrication system is further described in the documentation supplied with the engine.

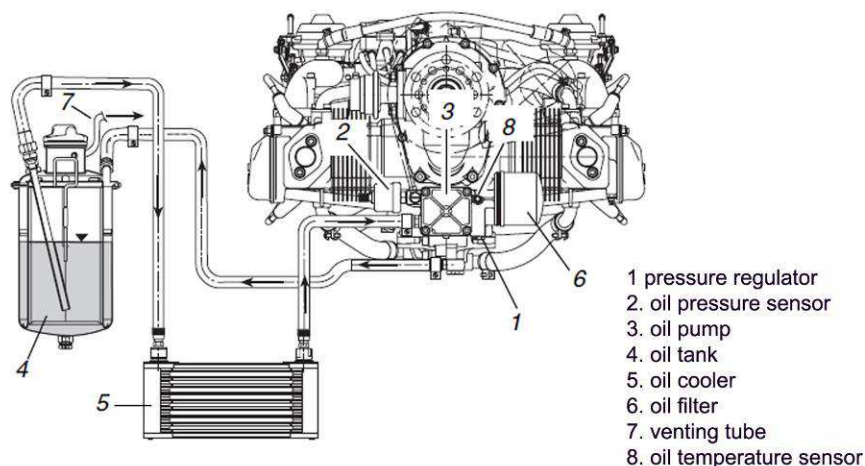


Figure 4-2 – Rotax Engine lubrication system

Cooling system description

The cooling system uses two forms of cooling. The cylinder heads are liquid cooled and the cylinders are ram air cooled. The radiator is located in the front of the lower engine cowl. The coolant is forced through the radiator by a water pump, driven from the crankshaft to the cylinder heads. From the top of the cylinder heads the coolant passes on to the expansion tank which allows for coolant expansion. The expansion tank is closed by a pressure cap with an excess pressure valve and return valve. When the temperature rises the coolant creates excess pressure, a relief valve opens and the coolant flows through a thin hose to the overflow bottle mounted on the firewall.

The engine cooling system is more completely described in documentation supplied with the engine.

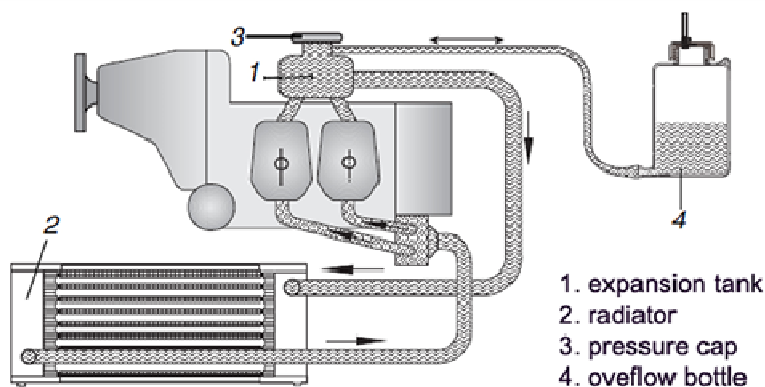


Figure 4-3 – Rotax Engine cooling system

4.2.2. Instructions for Authorized Maintenance, Repair or Alterations

4.2.2.1. Oil Filling

Refer to the Paragraph 1.6.2 on section 1.

4.2.2.2. Oil Emptying

- Tools needed: Phillips Screwdriver, Metric Wrench No. 17, a suitable oil can or sump.
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Unscrew the plug located on the bottom of the oil tank to empty out the oil.
To empty oil from the engine, pull out the return hose located on the bottom of the engine from oil tank (release hose clip first) and lower hose end to oil can.
Mount hose back.
It is recommended to empty oil when the engine is warm.
- Task proper accomplishment check: Verify if the hose located on the bottom of the engine from oil tank is properly connected.

4.2.2.3. Oil Filter Replacement

- Tools needed: Phillips Screwdriver, Metric Wrenches
- Parts needed: Oil Filter.
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Remove engine cowlings. If clearance is insufficient to remove filter. Remove the elbow on the left front (as viewed in flight direction) exhaust pipe. Loosen a clamp of that pipe on the exhaust muffler using wrench. Disconnect the elbow from the engine and turn the exhaust pipe slightly to move it from the oil filter.

Replace oil filter by a new one. See Maintenance Manual (Line Maintenance) for ROTAX Engine Type 912 Series for replacement instructions.

Connect the elbow to the engine and tighten the nuts slightly by fingers. Set exhaust pipe clearance from the radiator hose and oil filter. When clearances are set, tighten the elbow and clamp. Re-install the engine cowlings after oil re-filling.
- Task proper accomplishment check: None.

4.2.2.4. Coolant Filling

Refer to the Paragraph 1.6.3 on section 1.

4.2.2.5. Coolant Emptying

- Tools needed: Phillips Screwdriver, a suitable container for coolant.
- Parts needed: None.
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Remove upper engine cowl.
 2. Place suitable container under the engine.
 3. Check coolant quantity and fill some amount if needed.
 4. Disconnect the hose going from the radiator into the pump (on the lowest part of the cooling system)
 5. Empty coolant into a container.
 6. Mount hose back.
- Task proper accomplishment check: Verify if the hose going from the radiator into the pump is properly connected.

4.3. ENGINE BED

4.3.1. Description

The engine mounting bed is welded from chrome-molybdenum tubing and is attached to the firewall with 4 bolts. The bed is spring-mounted with four rubber vibration isolators.



Figure 4-2 – Engine Bed (1- bed, 2 - rubber vibration isolators)

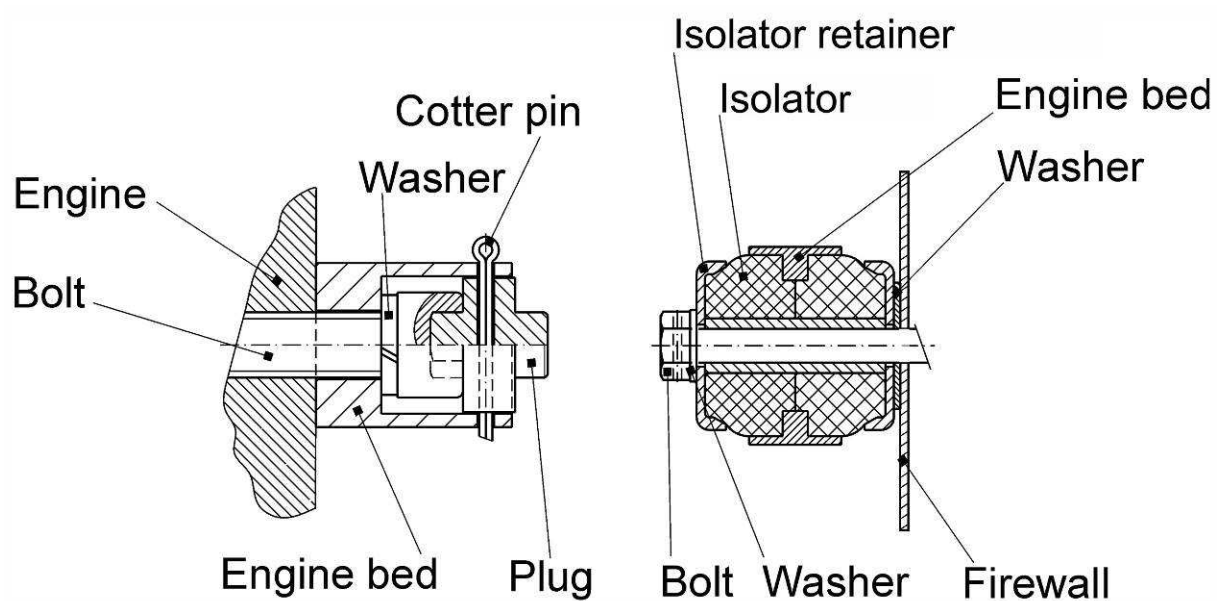


Figure 4-3 –Rubber vibration isolators details

4.4. ENGINE COWLINGS

4.4.1. Description

There are two fiberglass cowlings (upper and lower) which cover the engine. The interior sides of the cowlings is protected with fire resistant paint, the exterior is painted with aircraft paint. The upper cowling is connected to the lower cowling with quick-closing locks.

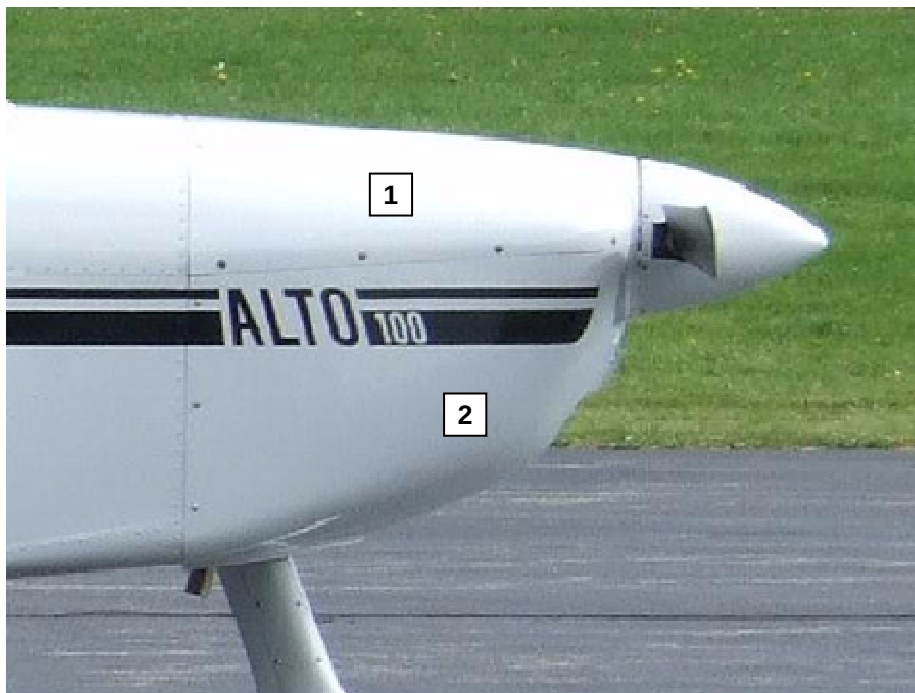


Figure 4-4 –Engine Cowlings (1- Upper Cowling, 2 – Lower Cowling)

1.4.2. Instructions for Authorized Maintenance, Repair or Alterations

1.4.2.1. Engine cowlings disassembly and assembly

- Tools needed: Phillips Screwdriver
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:
 - UPPER COWLING: The disassembly and assembly are both easy - just release the quick-closing locks. The upper cowling is usually removed during engine pre-flight inspection to check the engine compartment, operating fluids quantity (oil & coolant) and to check the engine installation.
 - LOWER COWLING: To remove it, unscrew the attachment screws connecting the cooler to the cowling face side, then remove two air hose (unscrew sleeves) and then unscrew the attachment screws connecting the cowling to the firewall flange.

The removed cowlings should be protected to prevent them from inadvertent damage. The cowling assembly is the reverse of disassembly.

- Task proper accomplishment check: Visually check if all quick-closing locks are locked.

Section 5 – FUEL SYSTEM

5.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft fuel system by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft fuel system.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

5.1.1. Description

Each wing contains an integral fuel tank with a capacity of approximately 46 liters (12.1 U.S. gallons) volume. Each tank is fitted with an air vent (output is under the wing) and a drain valve on the bottom side of the wing. Fuel flow from the tanks through the hoses to the fuel selector located on a central console under the instrument panel and then through a fuel filter to the electric and mechanical fuel pump then to the carburetors. The fuel return hose goes from the mechanical fuel pump into the right tank, which is considered as a "primary" tank. See figure below with the Scheme of fuel system.

The fuel tanks filler necks are placed on the upper side of the each wing. Fuel quantity is displayed by the engine monitoring system on the instrument panel.

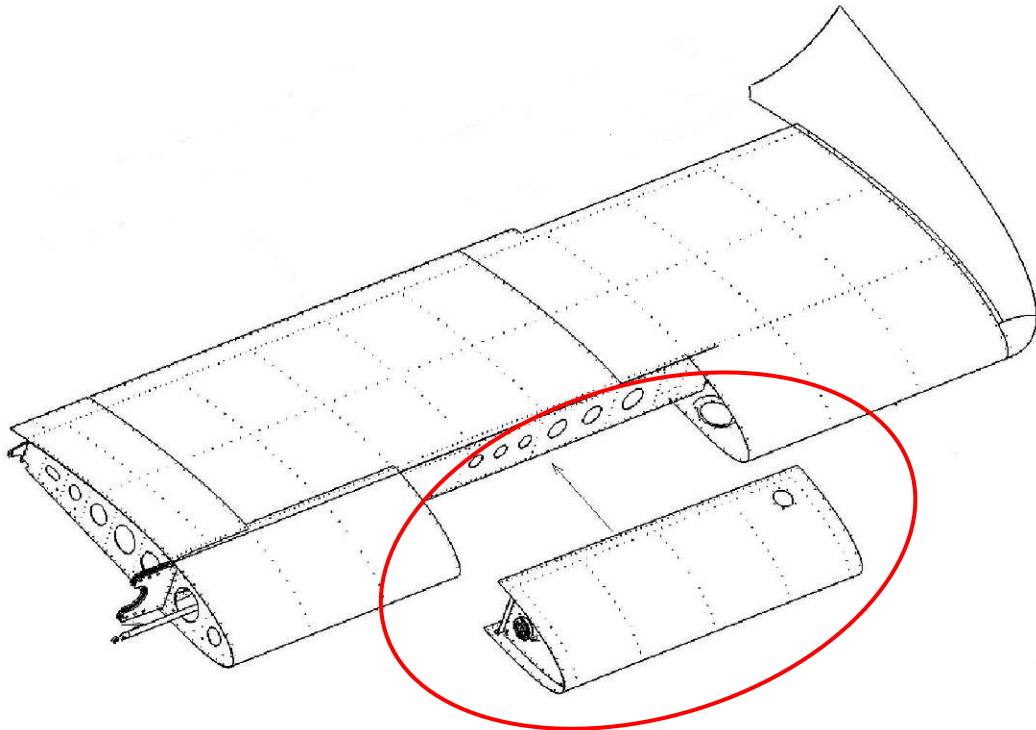


Figure 5-1 – Wing Tank

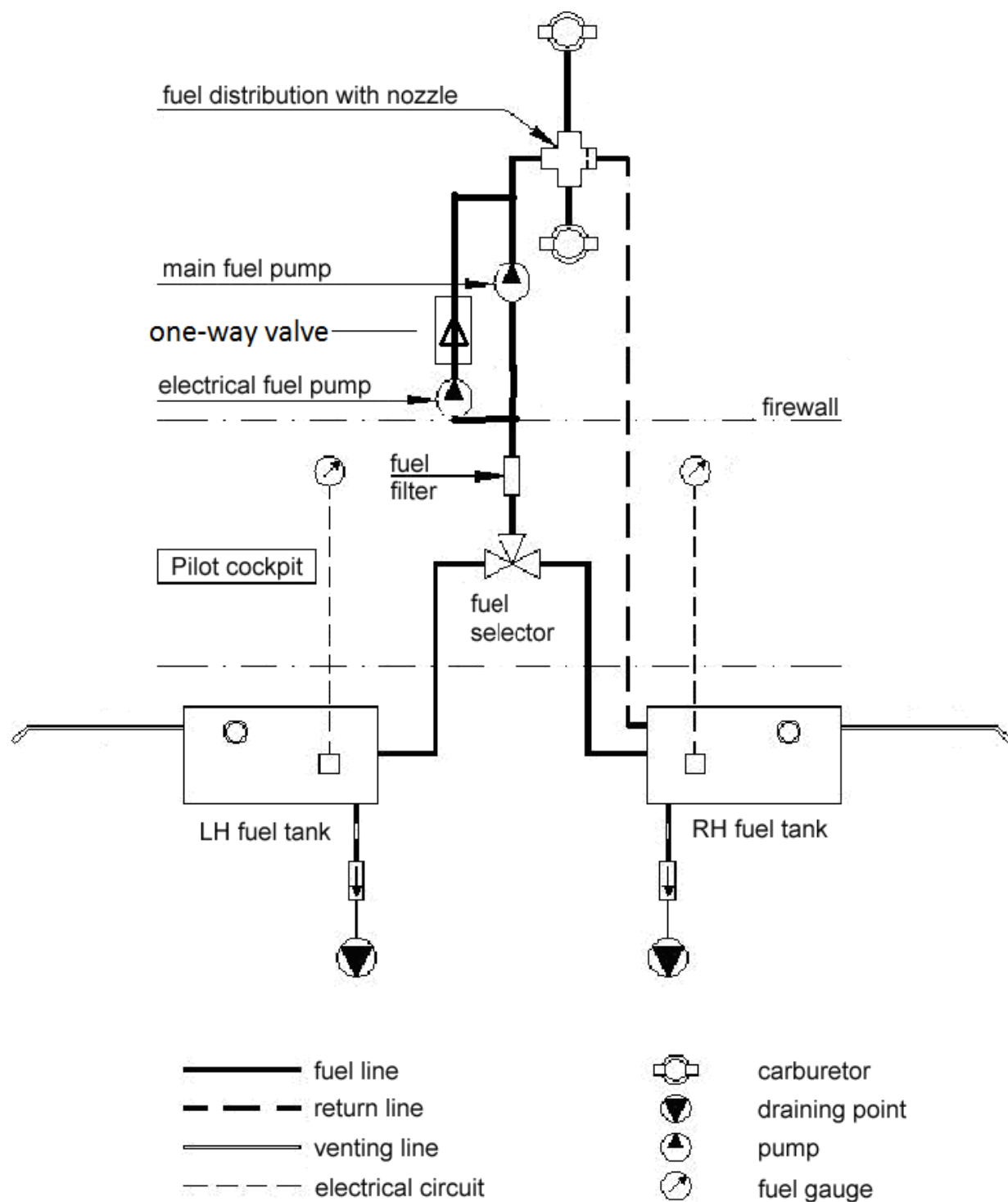


Figure 5-2 – Fuel system scheme.

5.1.2. Instructions for Authorized Maintenance, Repair or Alterations

5.1.2.1. Fueling

Refer to the Paragraph 1.6.1. on section 1.

5.1.2.2. Fuel tank draining

- Tools needed: None
- Parts needed: fuel tester or a transparent cup.
- Type of maintenance: Preventive Maintenance
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:
 1. Put the sump "fuel tester" or a transparent cup under the drain valve.
 2. Push open the drain valve.
 3. Drain required quantity of fuel
 4. Release the drain valve and visually check that there is no fuel leaking for the valve
 5. Repeat procedure for the opposite tank.



Figure 5-3 – Fuel drain valve.

- Task proper accomplishment check: Visually check, that the valve is closed and there is no leakage of fuel through it.

NOTE

Draining serves to eliminate impurities and deposits from the fuel.
Drain until clean fuel flows from the drain valve.

WARNING!

DO NOT SMOKE OR HAVE ANY OPEN FLAME DURING DRAINING!

5.1.2.3. Fuel Emptying

- Tools needed: 5/8 inch wrench
- Parts needed: a fuel container to be placed under the aircraft.
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Put the fuel container under the drain valve.
 2. Open fuel tank cap to speed up draining.
 3. Using the wrench, remove the drain valve.
 4. Drain the fuel. If it is necessary to drain all fuel, push airplane tail down (almost to the ground) to allow draining all of the unusable fuel.
 5. Reinstall the drain valve.
 6. Close fuel tank cap.
 7. Repeat procedure for the opposite tank..
- Task proper accomplishment check: Visually check, that the valve is closed and there is no leakage of fuel through it.

CAUTION

USE THE SAME PRECAUTIONS AS DURING FUELING.

WARNING!

DO NOT SMOKE OR HAVE ANY OPEN FLAME DURING DRAINING!

Section 6 – PROPELLER

6.1. GENERAL

Refer to the propeller manufacturer manual for proper maintenance.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft propeller.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

6.2. PROPELLER

6.2.1. Description

Manufacturer	Woodcomp S.R.O.
Model	Propuls 174/3/R
Type	Composite
Number of blades	3
Diameter	1740 mm (69 in)
Pitch	Ground Adjustable
Max RPM	2600

For more detail refer to the Woodcomp propeller manual supplied with your propeller.



Figure 6-1 – Woodcomp Propuls.

Section 7 – UTILITY SYSTEMS

7.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft utilities systems by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft utilities systems.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

7.2. HEATING

7.2.1. Description

A cockpit heating system is standard.

An air duct collects air from behind the water radiator via a fiberglass funnel. The air is then heated by flowing over the muffler shroud. The air then flows through another air duct which directs it through the butterfly valve and into the cockpit.

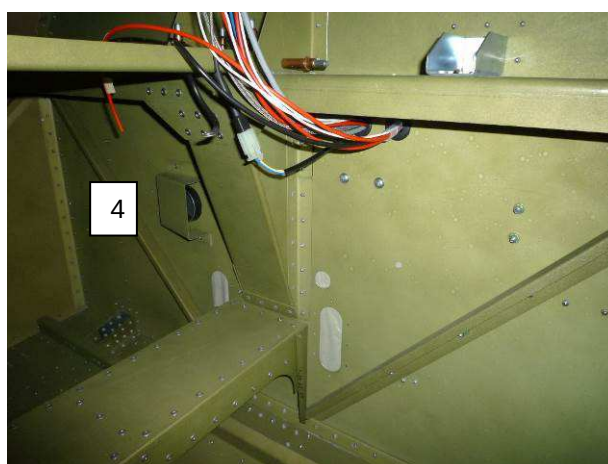
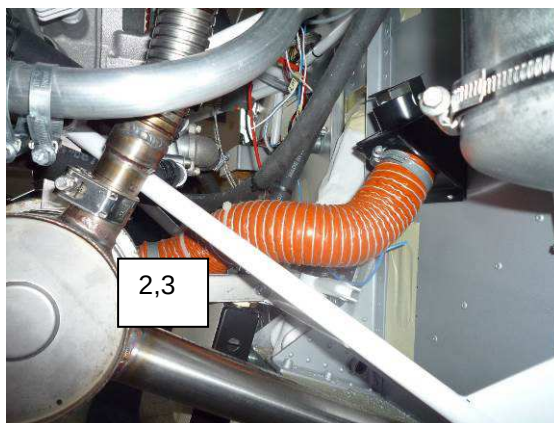


Figure 7-1 to 7-3 – Heating system

(1- inlet hose, 2-muffler shroud, 3- outlet hose, 4 – inlet of air into the cockpit)

7.2.2. Instructions for Authorized Maintenance, Repair or Alterations

7.2.2.1. Heating hoses replacement

- Tools needed: Screwdriver, metric wrenches
- Parts needed: hoses, clamps, wires
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Remove the engine cowlings, release hoses complexes, dismantle clamps
 2. Change hoses for new ones, install strip clamps
 3. Attach hoses
 4. Install the engine cowlings
- Task proper accomplishment check: Verify if all the clamps are firmly locked to the hoses.

7.3. VENTILATION SYSTEM

7.3.1. Description

Ventilation in the aircraft is provided by the sliding ventilation windows on each side of the canopy.

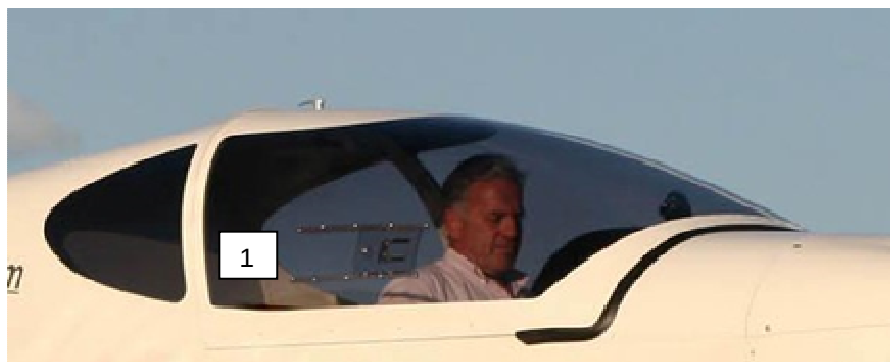


Figure 7-2 – Ventilation window (1- sliding window)

7.3.2. Instructions for Authorized Maintenance, Repair or Alterations

7.3.2.1. Sliding Window Substitution

- Tools needed: Screwdriver, metric wrenches
- Parts needed: spare window, spare rail
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. unscrew screws from plexi, take out the rails and window
 2. insert the window, insert rails
 3. screw up the screws - highly sensitive, tighten a screw without power.
- Task proper accomplishment check: Verify if the sliding windows are sliding freely.

Section 8 – INSTRUMENTS AND AVIONICS

8.1. GENERAL

Refer for the original Avionics or Instruments instructions for maintenance or checks. Replace defective instruments or avionics only with approved instruments.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft instruments and avionics.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

8.2. INSTRUMENTS AND AVIONICS

8.2.1. Description

The aircraft is equipped with an instrument panel containing flight, navigation, and engine instruments that are required for VFR operations. The Table below summarizes the equipment installed on the aircraft.



Figure 8-1 – Instruments and Avionics

INSTRUMENTS AND AVIONICS			
#	Instrument /Avionic	Brand	Model
1	Flaps Control	FlyBox	EFC-P
2	Vertical Speed Indicator	PRC	BC10-1B
3	Altimeter	Mikrotechna	LUN 1128
4	Airspeed Indicator	PRC	BK-15
5	Slip Indicator	Winter	WIQM2
6	Low Voltage Annunciator	Direct Fly	-
7	Fuel Pump Annunciator	Direct Fly	-
8	Trim Position Annunciator	Ray Allen Company	RP3
9	Left Tank Gauge	VDO	FL314
10	Right Tank Gauge	VDO	FL314
11	Engine Oil Temperature	Road	IH61.2B
12	Engine CHT	Road	IH71.2B
13	Tachometer	Road	IF61.2B
14	Engine oil pressure	Road	IE61.2B
15	Transponder	Garmin	GTX-327
16	VHF	Icon	IC-A210
17	GPS	Garmin	AERA 500
18	Compass	PRC	CM-24S

8.2.2. Instructions for Authorized Maintenance, Repair or Alterations

8.2.2.1. Installation and reinstallation of instruments - General

- Tools needed: Screwdriver, other tools depending on the instrument
- Parts needed: Instrument to be replaced or installed, for other parts, refer to manuals supplied with particular instruments
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:
 1. Dismantle the plexi canopy. Remove the top plate of the instrument panel to access the back part of the instruments.
 2. The installation procedure will depend on the instrument being installed. Follow the manufacturer recommendations (manuals supplied with particular instruments).
- Task proper accomplishment check: Refer to manuals supplied with particular instruments.

8.2.2.1. Installation and reinstallation of instruments – Pitot Static Instruments

- Tools needed: Screwdriver, other tools depending on the instrument
- Parts needed: Instrument to be replaced or installed, for other parts, refer to manuals supplied with particular instruments
- Type of maintenance: Heavy
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station

- Instructions:
 1. Dismantle the plexi canopy. Remove the top plate of the instrument panel to access the back part of the instruments.
 2. The installation procedure will depend on the instrument being installed. Follow the manufacturer recommendations (manuals supplied with particular instruments).
- Task proper accomplishment check: Refer to manuals supplied with particular instruments.

Section 9 – ELECTRICAL SYSTEM

9.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft Electrical System by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft fuel system.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

9.2. ELECTRICAL SYSTEM

9.2.1. Description

The electric system is a single-wire type with the negative side connected to the chassis. The power source is a single-phase generator integral to the engine. A 12V/7Ah maintenance free battery is located on the firewall.

The electrical system is controlled / protected by electric circuit breakers E.T.A installed on the instrument panel

The dual engine ignition is a separate part of the electrical system.

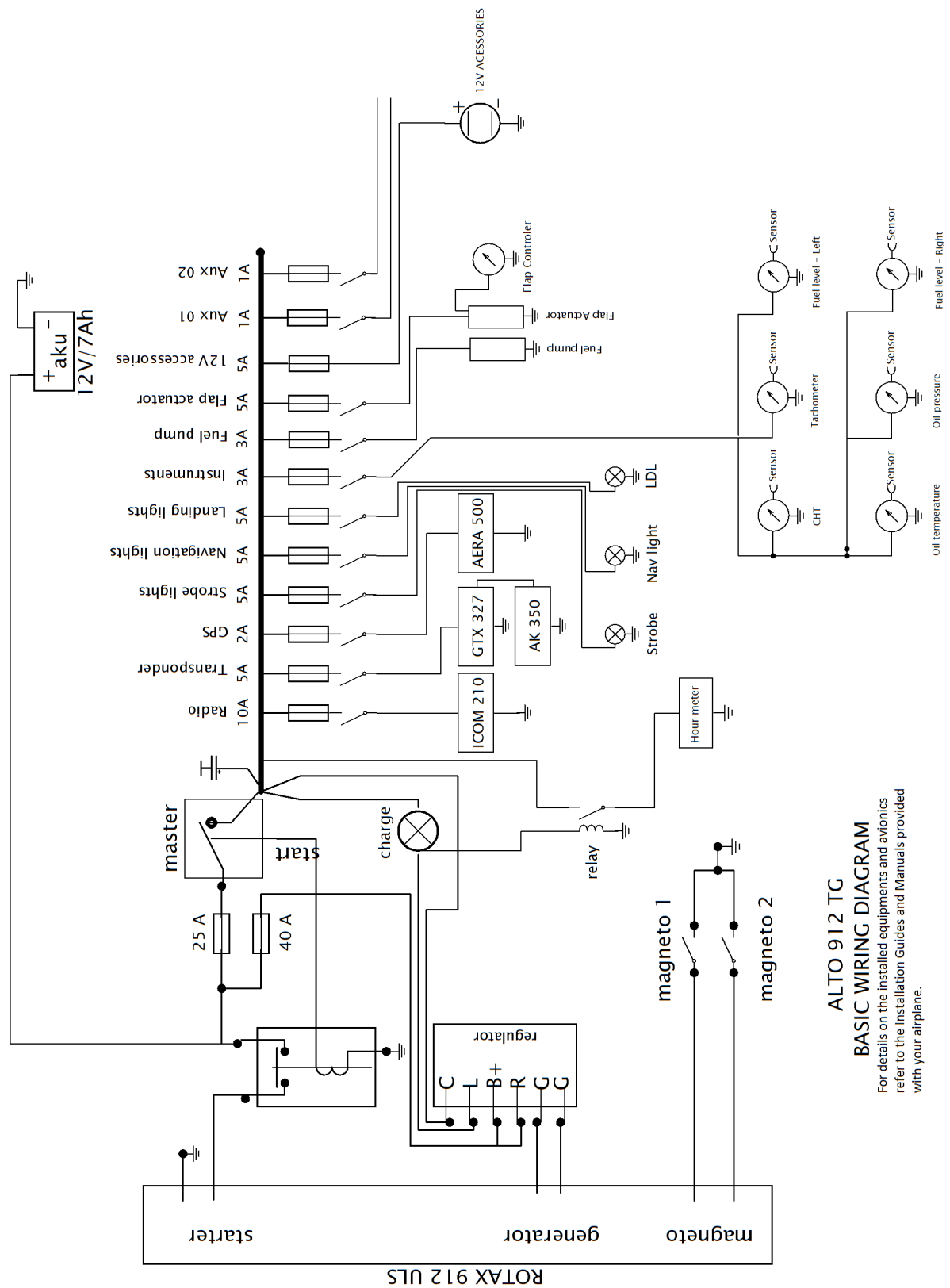
A 12V automotive socket located on the instrument panel is available.

Wiring diagram

The wiring system will vary and depends on the instrumentation, electronic equipment, and electric accessories of your aircraft.

The wiring diagram for the standard equipment installed on the Alto is shown on the picture below. For different configurations, the wiring diagram is available on the supplements of the POH (Pilot's Operating Handbook)

The specific wiring diagrams for the equipment installed on the aircraft are part of the instrument / avionics manuals, available on the POH Supplements.



ALTO 912 TG
BASIC WIRING DIAGRAM
For details on the installed equipments and avionics
refer to the Installation Guides and Manuals provided
with your airplane.

Figure -1 – Wiring diagram for the standard installation.

Circuit breakers/switches

The circuit breakers E.T.A are located on the lower edge of instrument panel

9.2.2. Instructions for Authorized Maintenance, Repair or Alterations

9.2.2.1. Inspection and operation checks of electrical system

- Tools needed: Screwdriver
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Aircraft owner (only for ELSA)
 - Sport pilot or higher
- Instructions:
 1. Remove top cover of the instrument panel and cowlings
 2. Inspect equipment, electrical assemblies, and wiring installations for damage, general condition, and proper functioning to ensure the continued satisfactory operation of the electrical system.
 3. A list of suggested problems to look for and checks to be performed are:
 - Damaged, discolored, or overheated equipment, connections, wiring, and installations.
 - Excessive heat or discoloration at high current carrying connections.
 - Misalignment of electrically driven equipment.
 - Poor electrical bonding (broken, disconnected or corroded bonding strap) and grounding, including evidence of corrosion.
 - Dirty equipment and connections
 - Improper, broken, inadequately supported wiring and conduit, loose connections of terminals, and loose ferrules.
 - Poor mechanical or cold solder joints.
 - Condition of circuit breaker and fuses.
 - Insufficient clearance between exposed current carrying parts and ground or poor insulation of exposed terminals.
 - Broken or missing safety wire, broken bundle lacing, cotter pins, etc.
 - Operational check of electrically operated equipment such as motors, inverters, generators, batteries, lights, protective devices, etc.
 - Voltage check of electrical system with portable precision voltmeter.
 - Condition of leds
- Task proper accomplishment check: None

9.2.2.2. Cleaning of electrical system

- Tools needed: Screwdriver
- Parts needed: None
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station

- Instructions:

Annual cleaning of electrical equipment to remove dust, dirt, and grime is recommended. Suitable solvents or fine abrasives that will not score the surface or remove the plating may be used to clean the terminals and mating surfaces if they are corroded or dirty. Only cleaning agents that do not leave any type of residue must be used.

Components must be cleaned and preserved in accordance with the manufacturer's instructions. Avoid using emery cloth to polish commutators or slip rings because particles may cause shorting and burning. Be sure that protective finishes are not scored or damaged when cleaning.

CAUTION**DISCONNECT BATTERY BEFORE CLEANING**

- Task proper accomplishment check: Operational check of electrically operated equipment such as motors, inverters, generators, batteries, lights, protective devices, etc.

9.2.2.3. Battery electrolyte corrosion cleaning /only for NON maintenance-free accumulators/

- Tools needed: Screwdriver, metric wrenches
- Parts needed: Brush, sodium bicarbonate solution
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:
 1. Remove upper cowling
 2. Remove the battery.
 3. Corrosion found on or near lead-acid batteries and cables can be removed mechanically with a stiff bristle brush and then chemically neutralized with a 10 percent sodium bicarbonate and water solution. After neutralizing, the battery should be washed with clean water and thoroughly dried.
 4. Reinstall battery.
- Task proper accomplishment check: Operational check of electrically operated equipment such as motors, inverters, generators, batteries, lights, protective devices, etc.

Section 10 – STRUCTURAL REPAIR

10.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft structures by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft fuel system.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

10.2. DESCRIPTION

10.2.1. Repair classification

Major repair, alteration or maintenance – Any repair, alteration or maintenance for which instructions to complete the task are excluded from the maintenance manual(s) supplied to the consumer are considered major.

Minor repair, alteration or maintenance – Any repair, alteration or maintenance for which instructions are provided for in the maintenance manual(s) supplied to the consumer of the product are considered minor.

Overhaul – Maintenance, inspection, repair or alterations that are only to be accomplished by the original manufacturer or a facility approved by the original manufacture of the product.

Repair Facility – A facility specifically authorized by the aircraft or component manufacturer to repair the product originally produced by that manufacturer.

10.2.2. Repair guidelines

NOTE

Repairs must be performed only by qualified persons – see Section 0 Paragraph 0.12.

Dents, cracks and loose rivets are the most frequent problems encountered during the operation of an all-metal airplane.

The following guidelines should be considered during repair:

1. The Strength in any cross-section of a repaired area/part should at a minimum be the same as the strength of the original undamaged area or part. The repaired area is not considered as a whole, but is considered as:
 - an original assembly unit
 - a stiffener or a patch

If the cross-section of a repaired area is (in all directions), at a minimum, the same as the one on the original part, or if a stiffener or a patch shape and design is similar, and the same materials are used, then there is no risk that the airframe will be weakened.
2. The load has to be transferred by a stiffener or a patch from one side of a repaired area to another. In some cases, the load transmission should be carried out at another structural member and then back to an undamaged part of the original structure.
3. The length of overlapping between a reinforcement (stiffener or patch) and the undamaged part of the original structure should be enough to assure a good joint. The length of overlapping is very important to assure no load concentration.

10.2.3. Damage classification

Various types of damage may occur during aircraft operation. It is important to correctly classify damage according to its character, size and especially, which part of the airplane has been damaged.

The important parts are the engine, engine mount, propeller, wing spar, wing box in the fuselage, elevator, control system, and landing gear.

Any damage must be repaired only by person/repair station qualified in accordance with Section 0,12 Qualification requirements. The damaged parts may be replaced in their entity by new ones supplied by the aircraft manufacturer.

Any damage and its repair should be recorded in the Log Book.

10.2.4. Materials used

10.2.4.1. Materials Specifications

Czech Standard		Z 424253.61
		Al Cu Mg 1
Mechanical properties (Minimum)	Yield Point Rp 0.2	260 MPa
	Tensile strength Rm	400 MPa
	Ductility [%]	15 (A10)
Chemical composition [%]	Al	rest
	Fe max	0.3
	Mn	0.3 - 0.9
	Cu	3.8 - 4.9
	Zn max	0.3
	Mg	1.2 - 1.8
	Si max	0.3
Half-finished products		by Al plated Sheets Al thickness max 10 mm 0.4 in
Equivalent Standards	U.S.A. Germany Russia	Alclad 2024 DIN 1745 D16AT

10.2.4.2. Materials Thickness

- 0.032 in (0.8 mm) Skin on forward part of fuselage
- 0.040 in (1.0 mm) Other small parts
- 0.050 in (1.2 mm) Wing Spar
- 0.063 in (1.6 mm) Center Section Spar
- 0.080 in (2.0 mm) Control Sticks
- 0.125 in (3.2 mm) Wing Spar
- 0.250 in (3.2 mm) Wing Spar
- 0.320 in (8.0 mm) Center Section Spar

10.2.5. Spare parts order

Use the form on Section 13, to order a replacement for a damaged or worn-out part. Do not hesitate to contact the airplane distributor (phone or fax) to request a required part.

On the form, additional information is necessary for systematic monitoring and analysis of our planes reliability.

10.2.6. Recommended reading

Many useful information you can find in the document:
Acceptable Methods, Techniques, and Practices – Aircraft Inspections, Repair & Alternations
AC 43.13-1B, AC 43.13-2A

10.3. INSTRUCTIONS FOR AUTHORIZED MAINTENANCE, REPAIR OR ALTERATIONS

10.3.1. SKIN REPAIR

- Tools needed: metalworker tools, drills, riveting pliers
- Parts needed: sheet metal, rivets
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station

- Instructions:

1 - To prevent small cracks (max. 0.2 in (5 mm)) from spreading, drill a hole of 0.06 in (1.5 mm), diameter at the end of the crack
2 - Small holes (up to Ø 0.16 in (4 mm)) can be repaired by a "POP" rivets

More extensive skin damage is repaired by means of a patch. Per standard repair procedures: Ref. AC 43.13-1B

Use a sheet of the same thickness as the repaired skin for the patch and frame.

A single row of rivets is used to rivet a small patch (patch area max. 7.75 sq in. 50 cm²). Use a double row for greater patches. Paint the internal side of the patch and frame before riveting and then paint the outside surface.

- Task proper accomplishment check: None

10.3.1.1. Riveting

- Tools needed: drills, riveting pliers
- Parts needed: rivets
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Avex / Blind rivets are used in most cases. Aircraft solid rivets are used only on wing spars, the wing box inside the fuselage and on some airplane parts with high loads. These parts must be repaired only by an authorized repair station.

When Avex / Blind rivets are to be removed, weaken the rivet head by drilling and then carefully drive out with a drift. Use a drill of the same size as the rivet. The rest of the rivet may be removed from the underside.

Avex / Blind rivet joints are sealed. If necessary use a suitable knife to cut out the sealing compound and then clean the joint. Warm up shortly the sealed joint using hot air gun 356-392°F (180-200 °C) to make it easier to cut.

In the following table lists the pop-rivets and characteristics:

AVEX Pop-rivets supplied by AVDEL Company									
1. material : Al Mg 2.5									
2. Shank material: steel, zinc-chromate coating surface									
Rivet No.:	Diameter		Length		Hole diameter		Max. riveting thickness		Head
	[in]	[mm]	[in]	[mm]	[in]	[mm]	[in]	[mm]	
1691-0410	0.126	3.2	0.3110	7.9	0.130-0.138	3.3-3.5	0.031-0.189	0.8-4.8	button
1691-0414	0.126	3.2	0.441	11.2	0.130-0.138	3.3-3.5	0.157-0.311	4-7.9	
1691-0512	0.157	4	0.374	9.5	0.161-0.170	4.1-4.3	0.047-0.248	1.2-6.3	
1691-0516	0.157	4	0.050	12.7	0.161-0.170	4.1-4.3	0.157-0.378	4-9.6	
1604-0412	0.126	3.2	0.374	9.5	0.130-0.134	3.3-3.4	0.094-0.248	2.4-6.3	counter-sunk

After drilling and de-burring the holes apply Adhesion "Emfimastic PU50" sealing compound or equiv. on either part. Use a spatula to make a homogenous layer (0.004 in, 0.1mm).

Riveting should be done before the sealing compound hardens (approximately half an hour).

For rivets spacing see surrounding skin; usually 0.8-1.6 in (20-40 mm) and keep a minimum distance of 0.3 in (8 mm) from the edge.

- Task proper accomplishment check: None

10.3.2. FIBERGLASS PARTS REPAIRS

10.3.2.1. Damage classification

Any damage of parts from reinforces plastics with epoxy matrix leads to increased saturation of the matrix with humidity and subsequently to loss of properties. Therefore we recommend carrying out their repair as soon as possible after the damage has occurred.

Two kinds of parts are made from these materials in the airplane:

- design appearance, non load-bearing parts (fairings)

According to the damage size we can divide repairs into:

- Small damage: surface defects, not affecting the stiffeners
- Medium damage: small cracks in fairings (excluding any defects to landing gear legs)
- Big damage: Major "Not repairable" structural, load-bearing parts (landing gear legs)

We recommend carrying out repairs by means of epoxy materials and glass stiffeners

10.3.2.2. General

Epoxy mixtures are prepared in a given mass ratio by means of weighing (accuracy of scales 1g).

10.3.2.2. Parts of external appearance

10.3.2.2.1. Small Damage

- Tools needed: sand paper, scissors,
- Parts needed: epoxy resin, drills, fiberglass cloth , degreasing agent
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

Repair of damage just by application of mastic and by varnish repair.

Preparatory grinding

For a good adhesion of repair layers it is necessary to carry out surface grinding at the utmost up to the depth of contact with the stiffener (do not damage). It is necessary to do surface grinding with overrun of 50 mm from the damage location smoothly to the top layer. It is suitable to do grinding with grain size of 160. Dry grinding equipment with suction from the grinding area is used. Al₂O₃ (fused corundum) can be used as grinding material.

Dust removing

It is made by wiping with clean and dry brush or by a vacuum cleaner.

Application of smoothing layer

After preparation of mixture (for material recommendation see tab below) and its eventual thickening to enhance the non curtaining capability (for vertical or lower areas) is performed its application onto the repair area by means of a spatula. For better distribution of deposited material on irregular surface it is possible to form it through the laid PE or PP transparent folia. After proper application the layer is without bubbles. Deposit thicknesses given by necessary thickness of surrounding layers (leveling) and ranges from 0.2 to 10 mm (0.0079 – 0.394 in) in one deposit.

Recommended materials for mixture preparation for application of leveling layer

Manufacturer, name	Type	Mixing ration	Dwelling	Rate of setting	Time of setting	Temperature	Fillers
HUNTSMAN, VANTICO, Araldite 2011 (AW106 + HV953U)	epoxy	A:B = 100:80	to be processed within 100min (for 0.1 kg mixture)	40%	24 hours (32hrs)	20 - 35°C (17 - 20°C)	thickening: Aerosil, micro balls

Grinding

Grinding or eventually sealing the surface is made after setting the mixture and possible tearing off the used folia. It is suitable to start grinding with grain size up to 160 and finish with grain size of at least 400. It is carried out by dry grinding equipment with suction from the grinding area. Al₂O₃ (fused corundum) can be used as grinding material. Basic material of the part must not be diminished at grinding.

NOTE

Especially in case of Carbon/Kevlar stiffener in the basic part, grinding through up to the stiffener must not occur (complication - see note about preparatory grinding at medium damage).

Finishing

See Section 11 – Painting and Coatings..

- Task proper accomplishment check: Visual check

10.3.2.2.2. Medium Damage

- Tools needed: sand paper, scissors,
- Parts needed: epoxy resin, drills, fiberglass cloth , degreasing agent
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station

- Instructions:

Repair of damage by replacing the stiffener part, by mastic and varnish repair. At such repair it is necessary to distinguish type of used stiffener (especially for grinding):

- C/K (Carbon/Kevlar), roving of fabric of black color (C) take turns with yellowish (K)
- G (glass), roving from milky white to transparent

Preparatory grinding

For good adhesion of repair layers it is necessary to do grinding up to the depth of damage. It is necessary to do surface grinding from the damage area with overrun at the least 25 mm (1 in) for every damaged stiffener layer smoothly up to the top layer and then about 50 mm (2 in) for finishing and mastic application. It is convenient to do grinding with grinding material having grain size of 160. It is carried out by dry grinding equipment with suction from grinding area. Al2O3 Aluminum Oxide Power (fused corundum) can be used as grinding material.

NOTE

In case of Carbon/Kevlar stiffener, Kevlar roving tend to rise up from the surface at grinding - it is difficult to grind them, we recommend using diamond grinding tool and one-way grinding.

Dust removing

Dust is removed by wiping with a clean and dry brush or by a vacuum cleaner.

Stiffener preparation

For this kind of repairs we recommend the stiffener G (glass) with plain weave, 150g/m², with surface protection (dewing) for epoxies.

Number of needed stiffener layers depends on depth of damage. It is possible to say that each layer of the mentioned fabric represents at proper saturation by matrix resin thickness of 0.5 mm (0.02 in).

Stiffener layers must be prepared (cut out) gradually from the smallest (the lowest) up to the bigger (upper), each with overrun of 20 mm (0.78 in).

Putting layers

After preparing lamination mixture (for recommendation of material see the Tab. on page 10-4), it is applied to the place of repair by means of rigid brush. The first stiffener is laid into the deposit and it is again saturated by brush. Another layer of stiffener is laid and saturated. When putting the last layer it is necessary to pay attention to a proper saturation and compression of stiffeners so that they cannot "come up" to the surface and subsequent useless damage at final grinding. For better saturation of the surface by resin and securing against curtaining it is possible to put PE or PP transparent foil across the surface. When applied properly, the layer is without bubbles. Repair thickness should exceed surrounding surface in this phase by about 0.5 to 1 mm (0.02 – 0.04 in) for finishing.

Grinding

Carry out grinding and eventually apply mastic on the surface after setting, eventually tearing off the folia (see small damage). It is suitable to begin grinding by abrasive with grain size of 160 and finish by grain size of at least 400. Dry grinding equipment is used with suction from the grinding area. Al₂O₃ can be used as grinding material. It is important not to diminish basic material of the part at grinding.

NOTE

Especially in case of Carbon/Kevlar stiffener in the basic part there must not be any grinding through up to the stiffener (complication -see the note about preparatory grinding).

Finishing

See Section 11 – Painting and Coatings.

- Task proper accomplishment check: Visual check.

10.3.2.2.3. Big Damage

At such damage we recommend to replace the affected part.

10.3.2.3. Structural Parts

- Instructions:

With these parts we do not recommend to do other repairs than small damage repairs as specified on Paragraph 10.6.2.2.1. In case of the other damage we recommend to contact manufacturer.

CAUTION

When repairing, it is necessary to carefully pay attention not to damage the stiffeners!

Cracks, permanent deformations and breaks are the most relevant type of damage.

Prior to undertaking the repair, clean the area to be repaired thoroughly, with soap and water. Remove any paint by wet or dry sanding. Superficial scars, scratches, surface abrasions, or erosion can generally be repaired by applying one or more coats of epoxy resin. The number of coats required (2 generally) depends upon the type of resin and severity of damage. The layers should have an overlap of 1.5-2 in (30-50 mm). After the resin has cured, sand off any excess and prepare the area for priming and painting.

Section 11 – PAINTING AND COATINGS

11.1. GENERAL

This Section identifies the line maintenance, minor repair, and minor alteration tasks that can be accomplished in the aircraft Painting and Coatings by the Aircraft owner (only for ELSA), a Sport Pilot or higher or a holder of an LSA repairman certificate with either an inspection or maintenance rating.

No heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, are authorized at this time, for this ALTO aircraft fuel system.

Any heavy maintenance, major repairs, or major alterations, as outlined in ASTM F2483 Sections 7 through 9, can only be performed by Direct Fly S.R.O. or only as provided by Direct Fly's' approved engineering data.

11.2. DESCRIPTION

11.2.1. Safety rules

When working with paints, thinners and solvents follow the following safety rules:

- it is necessary to follow safety rules for working with flammable and volatile substances
- working area must be properly aerated
- it is prohibited to smoke and anyway handle with free fire in a working area
- use protective working means such as goggles, gloves, respirator, etc.

11.2.2. Recommendation for paint repairs

CAUTION

By applying paints weight of airplane is increased and center of gravity position is changed. Increase in weight depends on type of coat and its thickness.

11.2.2.1. Washing and degreasing

It is possible to use both organic solvents and solvents based on water.

(a) **Organic solvents** - acetone, methylethyl-ketone (MEK), benzene, toluene, BASF Glasurit 360-4

- Applied by spraying on washed surfaces (e.g. mechanical sprayer, jet ejector) or by wiping with wet (by pouring, not by dipping because it would contaminate the whole volume of solvent) textile wad. After applying it, the agent is wiped off by clean absorbing material before solvent evaporation.
- **Advantages:** fast and reliable evaporation even from the corners and borders without additional warming.
- **Disadvantages:** it must be used without other dilution (expensive); not ecological (danger of water contamination); detrimental to health (must be carried out in an aerated area with personal protective means); waste (including dripping from the area) must be eliminated in the incinerating plants.
- **Use:** for Al-alloys surfaces, epoxy fiberglass

CAUTION

These agents must not be used for degreasing parts from plastics (PC - Lexan, PMMA - Plexiglas)

(b) **Water-based agents** - emulsion substances, wetting agents

- Applied also by spraying onto washed surfaces or by wiping with wet (by pouring and dipping) textile wad. After applying it, let it act for some time (see manufacturers recommendation) and then it is rinsed with clear water (by means of sponge dipped in ample amount of water or water jet).
- **Advantages:** (different according to the type of product: it is possible to highly dilute with water (cheap); ecological - waste (including dripping from the surface - due to ample amount of water it is necessary to contain it) can be generally eliminated after its additional dilution with water in public sewerage; the least detrimental to health.
- **Disadvantages:** slow and unreliable evaporating from corners and borders, additional warming (drying) mostly required - imperfect elimination of water results in wrong adhesion of paint coats; imperfect degreasing of fiberglass parts (not possible to use)
- **Use:** for Al-alloys surfaces and plastics

11.2.2.2. Bonding rivet heads, big irregularities and material transitions

CAUTION

In case that airplane surface remains without top coat, carry out just rivet head bonding.

After perfect degreasing carry out bonding of rivet heads, big irregularities and transitions of fiberglass parts with Al-sheet.

Epoxide and polyester bonding agents for car bodies are suitable; moreover for transitions between two types of material with increased elasticity. Recommended bonding agents are shown in the following tables.

Polyester bonding agents are applied with plastic spatula after being mixed with initiator. Grinding with grinding paper with grain size of 240 is made to smooth surface after drying. Remove dust after grinding and clean with degreasing agent.

Epoxide bonding agent is applied from the special jet with static mixer by means of extruding pistol. Excessive material is wiped off with spatula to final appearance before setting (slight recess is not a defect) - **do not grind!**

Recommended bonding agents:

Surface	Manufacturer, name	Type	Other components	Drying (grindable) [min] / 20°C
Al-alloys	BASF Glasurit 839-10 (base)	polyester	BASF Glasurit 948-36 (initiator)	30
transitions epoxide-fiberglass + Al-alloys	BASF Glasurit 839-45 (base)	polyester	BASF Glasurit 948-36 (initiator)	150
Rivet heads	3M DP-190 (base+ hardener)	epoxide (elastic)		360

11.2.2.3. Primer (paint)

In order to reach a uniform resistance to corrosion and smooth surface, we recommend carrying out application by means of spraying (air standard gun with the upper vessel, air HVLP gun, airless electric gun). The adjusting of the used gun (given by manufacturer) differs according to the type - air pressure, jet diameter. Primer should be applied in several sprayings (total thickness is not reached at a blow) with defined maximum dwell and total drying time till further treatment or handling.

Primer serves especially for anchoring (adhesion to the substrate) the top coats and can serve also for eliminating irregularities of the surface (function of filler, for grinding).

For surfaces from Al-alloys we recommend to use the etch-primers for light metals based on alkyd or materials based on epoxies or polyurethanes (2-component paints); specific recommendations according to the table below

Recommended primers

Surface	Manufacturer, name	Type	Further components	Surface mass [g/μm/m2]	Recommended thickness [μm]	Drying (between spraying/total) /20°C
Al-alloys	BASF Glasurit 801-1871 (base)	epoxies	BASF Glasurit 965-32/2 (hardener) Glasurit 965-50 (thinner)	1.62	25 (at 50 it can be also used as filler)	15min / 12hour

11.2.2.4. Bonding

After total drying of basic coat we recommend to carry out total bonding of irregularities including repairs of bonding. Recommended binders are shown in the table - Recommended bonding agents.

After drying perform grinding with emery paper with grain size of 240 until the surface is smooth. After grinding clean dust and wipe off with grease remover and perform repairing paint coat by primer (1/3 of coat thickness)

11.2.2.5. Top coat

In order to reach smooth surface we recommend again carrying out the paint coat by spraying (see paragraph 5.6.2.3).

Top coat serves especially for creating the coat resistant to weather and external effects for aesthetic rendering of the unit. Considering the higher loading by external effects we recommend to use top materials, exclusively two-component ones, on the acrylic-polyurethane or polyurethane basis, always with guaranteed adhesiveness to the used base coat (according to manufacturer). It is possible to use some of the coat systems for car repairs. In our recommendation (see the table below) there are two types of colors: single coat (color shade and protection in one), double coat (one-component color shade is formed by the substrate and protection is ensured by two-component transparent top coat). By single coat paint coat it is possible to reach the wide spectrum of colors, but it is difficult to do metallic paint coats (we do not recommend them).

Type of colors	Manufacturer name	Type	Other items	Surface weight [g/μm/m ²]	Recommended thickness [μm]	Drying time (between spray-coats / total) /20°C
single coat	BASF Glasurit R-18/ shade (base)	acrylic-polyurethane	BASF Glasurit 922-18 (standard hardener) Glasurit 352-91 (standard thinner)	1.0-1.6 (by shade)	20-40	15 min / 16 hrs
Double-coat	base color coat: BASF Glasurit R-55/ shade (base)	acrylic-polyurethane	BASF Glasurit 352-216 (thinner, long)	1.2-1.4 (by shade)	15-20	10 min / 20 min
	top coat, bright:: BASF Glasurit 923-155 (base)	acrylic-polyurethane	BASF Glasurit 929-73 (standard hardener) Glasurit 352-91 (standard thinner)	0.92	30-40	5 min / 5 hrs

11.3. INSTRUCTIONS FOR AUTHORIZED MAINTENANCE, REPAIR OR ALTERATIONS

11.3.1. SMALL DAMAGE

Small damage is a deterioration of corrosion resistance. At repair the situation is made more difficult by the fact that the substrate for repair coats is not a compact surface of basic material but mostly all coats of surface protection (after grinding), of which not all are suitable for (in ageing stage) for good adhesion of paint coats. Therefore we recommend carrying out such repairs by a verified system.

Before repairing it is necessary to differentiate the type of the existing top coat, single coat and two-coat (with the top coat). For repair it is necessary to follow the used type of color.

It is suitable to choose the delimited area (e.g. connection of sheets, wing edge) for the scope of the place which is being repaired transition is then better blended. In the case that it is not possible to choose the area in this way, it is necessary to take into consideration the higher difficulty of the procedure as for the uniformity of shade and elaboration of coat transition

- Tools needed: abrasive-coated paper
- Parts needed: materials for primer and for top coat
- Type of maintenance: Line
- Level of Certification needed:
 - Repairman (LS-M) or Mechanic (A&P) – both with the minimum level of certification to perform heavy maintenance of SLSA in the U.S.
 - FAA approved Part 145 Repair station
- Instructions:

1 – Grinding:

For good adhesion of the repair coats it is necessary to carry out grinding of the old paint coat at least up to such depth as the depth of damage. With two-coat type of the top coat it is necessary to add at least 50 mm (2 in) for the run-out of the top coat. Grinding can be started with abrasive having grain size of max. 160 and to finish with grain size of 400. It is made by the grinding equipment with the suction from the area of grinding or by manual grinding under water.

CAUTION

Anodized coat must not be destroyed by grinding on the Al-alloy sheet.

2 – Degreasing:

Refer to paragraph 11.2.2.1

3 – Application of primer:

For reaching the satisfactory equal adhesion we recommend to carry out a spray coat of the place to be repaired by adhesive interlayer (see the tab. below).

Adhesive interlayer

Surface	Manufacturer, name	Type	Surface weight [g/μm/m ²]	Recommended thickness [μm]	Drying (total) /20°C
Al-alloys	BASF Glasurit 934-0	single- component	0.8	5-10	max. 15 min
Epoxy fiberglass					
Old paint coats					

Subsequently the primer is applied according to the table Recommended primers. Paint coat thickness is given by necessary thickness of surrounding coats (leveling).

CAUTION

In case that the primer was not removed by the previous step, it is not necessary to apply the primer again. The original ground primer with adhesive intercoat is enough.

Actual application of primer will be carried out in the same way as for the total spray-coat (see paragraph 11.2.2.3).

4 – Application of Top Coat:**CAUTION**

For repairing it is necessary to choose the identical type (single coat, double coat) of the repair color as on the original surface.

Application of the top coat will be carried out by spraying as for the total spray coat (see 5.6.2.3) with the exception of used thinners and hardeners. Due to the need of smooth transition to the basic surface and it is necessary to use so called “spraying into the surface” using longer time of drying initiations for a good result of work. The recommended material is shown in the table on paragraph 11.2.2.5.

- Task proper accomplishment check: Visual check.

Section 12 – REVISIONS

12.1. GENERAL

When necessary, the revisions or amendments to these instructions shall be issued in the form of bulletins with attached new pages.

This Section will also be revised in all revisions of this Aircraft Maintenance Manual.

12.2. REVISION NOTES

[illegible]

Section 13 – FEEDBACK FORM

13.1. GENERAL

The owner/operator of a LSA is responsible for notifying the manufacturer of any safety of flight issue or significant service difficulty upon discovery. The following form may be used & sent to Direct Fly S.R.O by email or fax to the addresses listed on Paragraph 0.1 of Section 0.

DF				
1 Notice of owner/operator about operating failure of the aircraft				
Dear customer,				
We would like to ask you for your assistance at obtaining information for continuous increasing reliability of airplanes produced by our company. Please fill out the card in case of any failure on your airplane. After filling out, send it to the address mentioned above. By sending us the card, you provide us with valuable data, which enable us to improve reliability of your airplane as well.				
1.1 Mandatory data				
Airplane type:		Registration mark:		Airplane S/N:
1.	Flight units: flight hours – number of landing		: -	
2.	Failure detection date: day – month – year (format: "dd mm yy")			
3.	Failure has been detected at: (encircle appropriate number)	1. Flight 2. Taxiing 3. Take-off run 4. Take-off 5. Touch down 6. Landing run 7. Daily inspection 8. Periodical inspection 9. Other		
4.	Consequences for operation: (encircle appropriate number)	1. No consequences 2. Airplane put out of operation 3. Airplane returned from take-off 4. Flight with damaged aggregates 5. Emergency landing 6. Occurrence on the ground 7. Other		
5.	Failure description:			
6.	Identification of the damaged part			
	Failed part name	Catalogue number of failed part	Worked out operation units	Serial No.*
	Note: State maximum detectable data		* fill out if required	
7.	Notes, additional data (kind of failure, defect, incorrect activity):			
1.2 Additional data				
8.	Claimed (encircle what applicable): YES - NO		No. of claim:	
9.	Order of spare parts (encircle what applicable): YES - NO			
	Item	Name of part	Catalogue part No.	No. of pieces
	Owner:		Home airport:	
	Operator:		Date:	Elaborated by:

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